5 Kidney Metabolism

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CHAPTER OUTLINE

KEY POINTS

- The kidney has one of the highest specific metabolic rates among all organs. At the level of the whole organ, the major sources of energy for the kidney include lactate, fatty acids, and glutamine.
- Different nephron segments may prefer different substrates for the adenosine triphosphate production, but much remains to be learned about substrate utilization and metabolic activities in specific nephron segments and cell types in the kidney in vivo.
- The distinct metabolic features of nephron segments may be in part linked to the gradual decrease in tissue oxygenation level from the cortex to the inner medulla.
- Most of the energy consumed by the kidney is used to fuel the reabsorption of sodium and, secondarily, other solutes. Greater efficiency of energy utilization is achieved in some nephron segments, such as the proximal tubule, by leveraging passive reabsorption of sodium.
- In addition to its fueling function, renal metabolism may have an important regulatory function where metabolic intermediaries or activities regulate signaling pathways and gene expression that are not directly related to bioenergetics.

The kidney has one of the highest specific metabolic rates among all organs. The specific metabolic rate in the kidney of a human, which has been estimated to be more than 400 kcal/kg tissue/day, is as high as the heart and higher than other organ tissues at rest.[1] Although the term metabolism may be used to refer to all chemical reactions that occur in an organism, the term is used more narrowly in this chapter, interchangeably with intermediary metabolism. Intermediary metabolism, which refers to intracellular processes that convert nutritive substances to energy, cellular components, and waste products, is essential for fueling the physiologic function of the kidneys, especially renal tubular transport. Changes in renal energy and substrate metabolism may influence tubular transport by changing adenosine triphosphate (ATP) availability.

In addition to its fueling function, intermediary metabolism may have important regulatory roles in which metabolic pathways and intermediate products influence gene expression, signal transduction, and other regulatory pathways in the kidneys (Fig. 5.1).[2,3] Alterations in intermediary metabolism and related cellular functions such as mitochondrial function play an essential role in the development of acute kidney injury and chronic kidney disease,[4,5] as well as the development of disease conditions that intimately involve the kidneys such as hypertension.[3] Therefore knowledge of kidney metabolism is important for understanding both normal kidney physiology and the development of kidney-related disease and may provide new targets for the prevention or treatment of kidney-related disease.

This chapter provides an overview of the basics of metabolism and unique metabolic features of the kidney, summarizes the current understanding of kidney metabolism on the basis of studies of intact kidneys in vivo

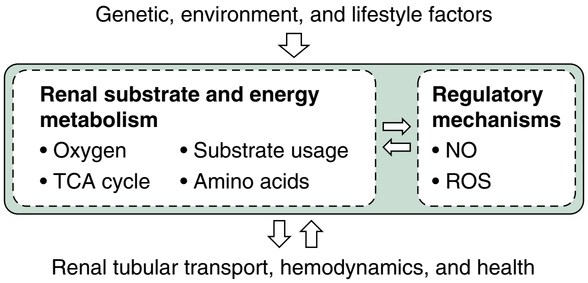

Fig. 5.1 Renal metabolism may influence renal physiology, health, and disease through energy production or regulatory mechanisms not directly related to energy production. *NO,* Nitric oxide; *ROS,* reactive oxygen species; *TCA cycle,* tricarboxylic acid cycle.

and isolated renal tissues describes the relation between kidney metabolism and tubular transport and discusses the regulatory role of intermediary metabolism in the kidney that is beyond bioenergetics. This chapter focuses on normal kidney metabolism. Some examples of abnormal kidney metabolism will be mentioned, but the role of kidney metabolism in the development of disease is covered in greater detail in chapters that deal with specific diseases.

OVERVIEW OF KIDNEY METABOLISM

KIDNEY STRUCTURE AND FUNCTION

The structure and function of the kidney are described in detail in Chapter 2. A brief summary is provided here to serve as the context for understanding kidney metabolism. The structure and function of the kidney are highly compartmentalized. The primary functional unit of the kidney is the nephron. The number of nephrons averages approximately 1 million in a human kidney. Each nephron consists of a glomerulus and a Bowman capsule connected serially to a proximal tubule, a loop of Henle, and a distal tubule. Several nephrons drain into a shared collecting duct. Tubular segments have distinct transport characteristics. The glomerulus filters the plasma in bulk, and 98% of the filtered water and sodium is reabsorbed by the tubule. The afferent arteriole feeds blood to the capillary tuft in the glomerulus, while the efferent arteriole drains the glomerulus and then forms a second capillary network that surrounds the tubule before returning blood to the veins. Juxtamedullary nephrons, of which the glomeruli are in the deeper region of the renal cortex close to the renal medulla, have long loops of Henle that extend to the renal inner medulla. Special capillaries called vasa recta run along these long loops of Henle, an anatomic arrangement that is essential for urinary concentration.

SUBSTRATE METABOLISM

Substrates enter the kidney by renal blood flow (RBF) and glomerular filtration and enter renal epithelial cells by substrate transporters, often facilitated by the inward-directed Na^+ gradient created by the sodium pump, as discussed thoroughly in Chapter 8. Oxygen is likewise delivered by RBF to the epithelial cells. Once in the cell, substrates face one of three fates: 1. transport across the epithelium back into the blood (reabsorption); 2. conversion into another substrate (e.g., lactate to pyruvate); or 3. oxidization to CO_2 in the process of cellular ATP production.[6]

A detailed description of the biochemistry of fuel metabolism can be found in biochemistry textbooks. The aspects of fuel metabolism that are relevant to the kidney are summarized here to provide a foundation for detailed discussions of kidney metabolism in later parts of the chapter. A variety of substrates may be used as fuel in the kidney (Fig. 5.2). Long-chain free fatty acids are modified to form acyl-coenzyme A (CoA) in the cytosol, transported into mitochondria via the carnitine shuttle, and converted to acetyl-CoA via β oxidation. Fatty acids with shorter chains, including those produced from very long chains by peroxisomes, may diffuse into mitochondria without using the carnitine shuttle. Acetyl-CoA reacts with 4-carbon oxaloacetate to form 6-carbon citrate. Citrate is converted to a series of metabolic intermediates in the tricarboxylic acid (TCA) cycle, during which reduced nicotinamide adenine dinucleotide (NADH) and reduced flavin adenine dinucleotide (FADH2) are produced. Fatty acid metabolism produces ketone bodies, especially β-hydroxybutyrate, which may also be used to generate acetyl-CoA and enter the TCA cycle in the kidney. Glucose is metabolized to pyruvate through glycolysis, which produces two net ATP and two NADH. Pyruvate may be converted to lactate or enter the TCA cycle in the form of acetyl-CoA. Lactate may be converted back to pyruvate, which may either feed the TCA cycle or generate glucose via the gluconeogenesis pathway. The latter may form part of the Cori cycle. Glutamine contributes to ATP production when it is converted to glutamate and subsequently the TCA cycle intermediate α-ketoglutarate. Other amino acids may be catabolized to produce acetyl-CoA and other metabolites and enter the TCA cycle. Some circulating TCA cycle intermediates also may be taken up by the kidneys and used as fuel.

ATP PRODUCTION

Renal epithelia, except in the descending and thin ascending limbs of the loop of Henle, are packed with mitochondria (see Chapter 2). All the pathways of fuel oxidation take place in the mitochondrial matrix, except for glycolysis, which occurs in the cytosol. Substrates in the cytosol can freely cross the outer mitochondrial membrane through integral membrane porins. These substrates, as well as adenosine diphosphate (ADP) and phosphate (the building blocks of ATP), cross the inner mitochondrial membrane into the mitochondrial matrix via specific substrate transporters driven by their respective concentration gradients or by the H^+ gradient created by the electron transport chain (ETC; Fig. 5.3).

As illustrated in Fig. 5.4, amino acids, fatty acids, and pyruvate are metabolized to acetyl-CoA and enter the TCA cycle. With each turn of the cycle, three molecules of NADH, one molecule of FADH2, one molecule of guanosine triphosphate (GTP) or ATP, and two molecules of CO_2 are released in oxidative decarboxylation reactions (Table 5.1). Electrons carried by NADH and FADH2 are transferred into the mitochondrial ETC, a series of integral membrane complexes located within the inner mitochondrial

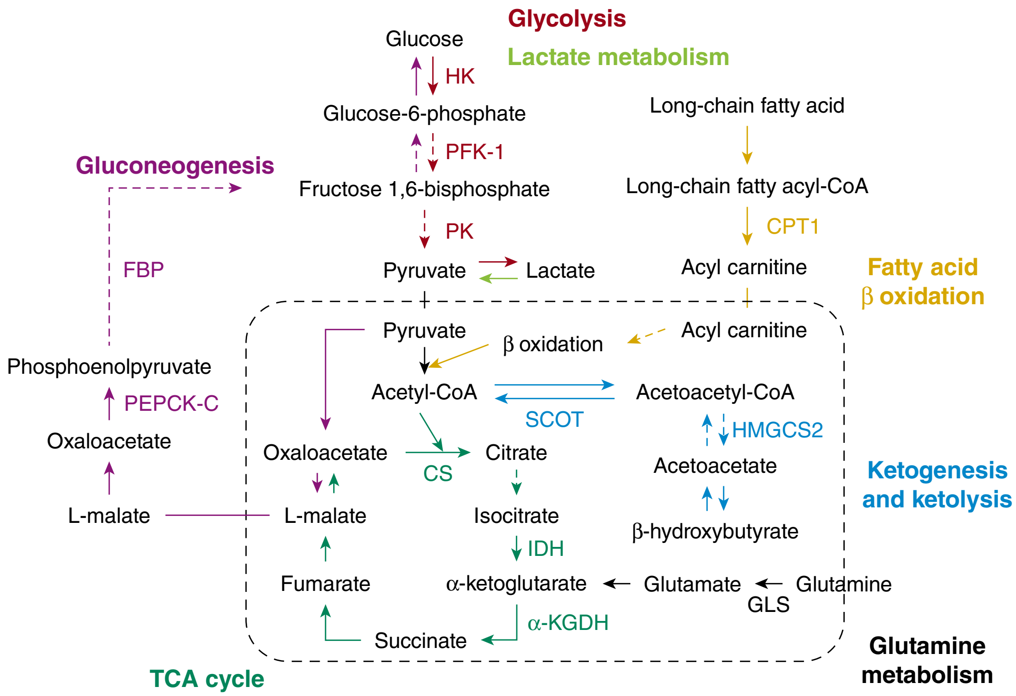

Fig. 5.2 Major substrate metabolic pathways in the kidney. To make the figure legible, only rate-limiting or key enzymes are shown, and *arrows* that contain multiple enzymatic steps are shown as *dashed arrows.* The box surrounded by the *dashed line* indicates mitochondria. *α-KGDH,* α-Ketoglutarate dehydrogenase; *CPT1,* carnitine palmitoyltransferase I; *CS,* citrate synthase; *FBP,* fructose 1,6-bisphosphatase; *GLS,* glutaminase; *HK,* hexokinase; *HMGCS2,* mitochondrial HMG-CoA synthetase; *IDH,* isocitrate dehydrogenase; *PEPCK-C,* cytosolic phosphoenolpyruvate carboxykinase; *PFK-1,* phosphofructokinase-1; *PK,* pyruvate kinase; *SCOT,* succinyl-CoA:3-ketoacid-CoA transferase. (From Tian Z, Liang M. Renal metabolism and hypertension. *Nat Commun.* 2021;12:963.)

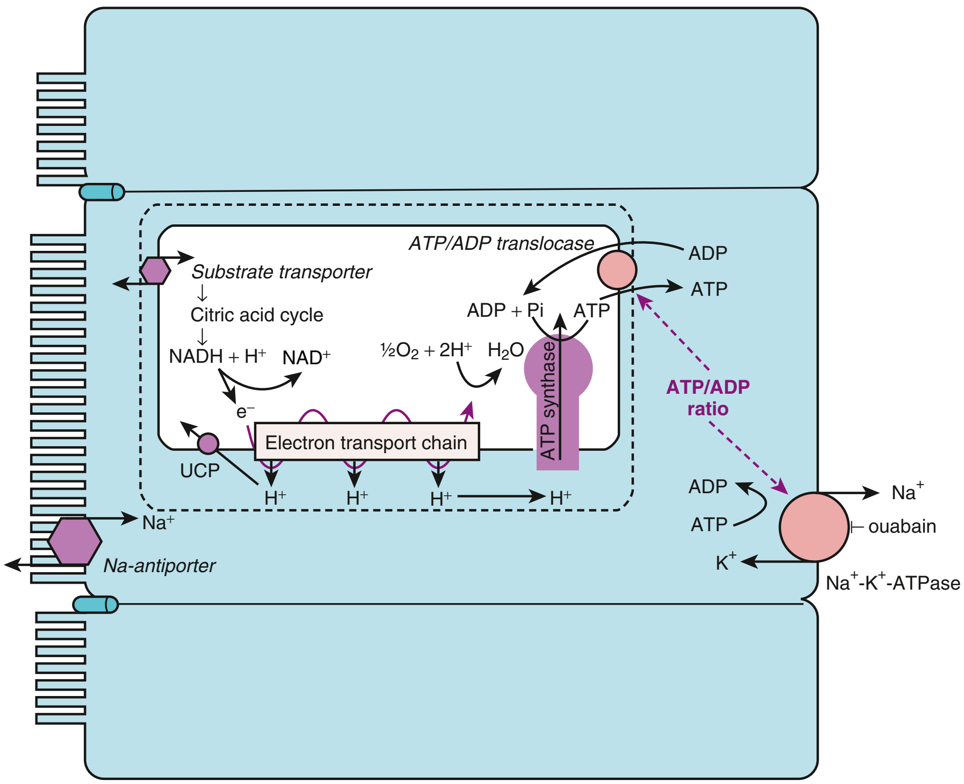

Fig. 5.3 Whittam model. Coupling of adenosine triphosphate (ATP) utilization by Na^+-K^+-ATPase to ATP production by mitochondrial oxygen consumption (Q_{O_2}). Hydrolysis of ATP produces ADP plus inorganic phosphate *(Pi),* which lowers the ATP/ADP ratio, a signal to increase ADP uptake into the mitochondria and increase ATP synthesis.

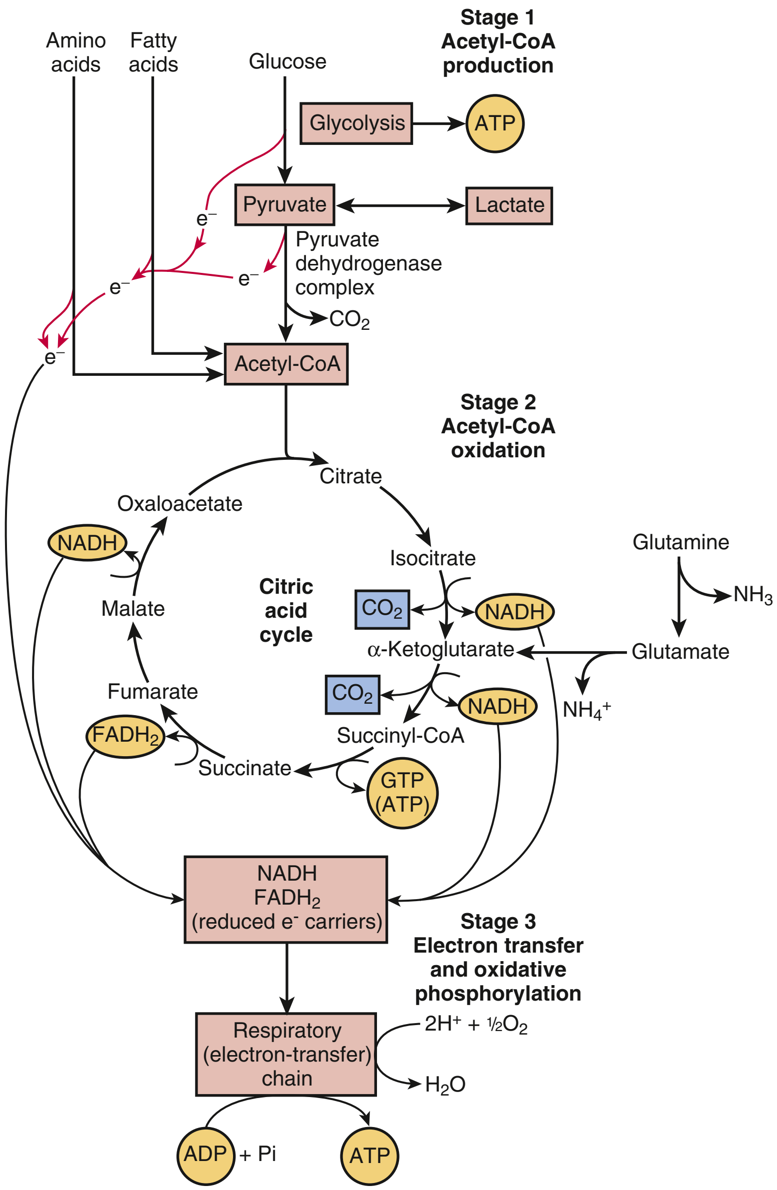

Fig. 5.4 Catabolism of proteins, fats, and carbohydrates in three stages of cellular respiration. Stage 1: oxidation of fatty acids, glucose, and some amino acids yields acetyl–coenzyme A (*CoA*). Stage 2: oxidation of acetyl groups in the citric acid cycle includes four steps in which electrons are abstracted. Stage 3: electrons carried by reduced nicotinamide adenine dinucleotide (*NADH*) and reduced flavin adenine dinucleotide (*$FADH_2$*) are funneled into a chain of mitochondrial (or, in bacteria, plasma membrane–bound) electron carriers (the respiratory chain) that ultimately reduces O_2 to H_2O. This electron flow drives the production of ATP. Also indicated are two proximal tubule pathways: 1. oxidation of lactate through pyruvate and acetyl-CoA and 2. glutamine conversion to glutamate and α-ketoglutarate in the mitochondria with the production of two molecules of NH_3, which is the main source of NH_3 secreted during acidosis. (Modified from Nelson DL, Cox MM. *Lehninger Principles of Biochemistry*. 5th ed. New York: WH Freeman; 2008.)

membrane, where the electrons are sequentially transferred, ultimately to oxygen, which is reduced to H_2O. NADH and FADH2 oxidization provoke the transport of H^+ from the matrix to the inner mitochondrial space.

The release of the potential energy stored in the H^+ gradient across the inner mitochondrial membrane provides the driving force for ATP synthesis from ADP by the ATP synthase: H^+ is transported into the matrix coupled to the production of ATP from ADP and inorganic phosphate (Pi) (see Fig. 5.3). The newly synthesized ATP is extruded from the matrix into the intermembrane space via the ADP–ATP countertransporter known as adenine nucleotide translocase and then exits the mitochondria across the permeable outer membrane. In the cytosol, ATP is available to bind to ATPases such as plasma membrane Na^+-K^+-ATPase. Complete oxidation of palmitate, a 16-carbon fatty acid, produces up to 129 ATP, and a molecule of glucose produces up to 38 ATP if it is metabolized completely through the TCA cycle. The actual number of ATP molecules produced is lower because of several reasons.

The oxidation of substrates is coupled to ATP synthesis by an electrochemical proton gradient. This coupling can be influenced by uncoupling protein isoforms (UCPs) located in the mitochondrial inner membrane and expressed in

Table 5.1 Adenosine Triphosphate (ATP) Yield From Metabolism of 1 Glucose Molecule

Process	Direct Product	Final ATP
ATP Yield from Complete Oxidation of Glucose		
Glycolysis	2 NADH (cytosol)	5[a]
	2 ATP	2
Pyruvate oxidation (2 per glucose)	2 NADH (mitochondrial matrix)	5
Acetyl–coenzyme A oxidation in citric acid cycle (2 per glucose)	6 NADH (mitochondrial matrix)	18
	2 FADH2	4
Total yield per glucose		30
ATP Yield from Glycolysis of Glucose		
Glycolysis	2 ATP, 2 NADH	2

a tissue-specific manner. UCPs create a proton leak that dissipates the proton gradient available to drive oxidative phosphorylation (see Fig. 5.3). It has been reported that UCP-2 is expressed in the renal proximal tubule and thick ascending limb (TAL) (not in glomerulus or the distal nephron) and that its expression is elevated in kidneys of diabetic rats.[7]

UNIQUE OVERALL FEATURES OF KIDNEY METABOLISM

In addition to a high specific metabolic rate, kidney metabolism is unique in that most, possibly 80% or more, of the oxygen consumed by the kidney is used to support active transport, primarily by fueling Na^+-K^+-ATPase. Another prominent feature of kidney metabolism is that blood flow and tissue oxygenation vary substantially between kidney regions. The high blood flow in the renal cortex exceeds the tissue's metabolic needs but is necessary for the bulk filtration at glomeruli that is essential for removing whole-body metabolic wastes. The partial pressure of oxygen (PO_2) is about 50 mm Hg in the renal cortex. Tissue PO_2 decreases gradually into the renal medulla, reaching 10 to 15 mm Hg in the renal inner medulla. These features are important for understanding kidney metabolism and its relation with kidney function and disease as detailed later in this chapter.

KIDNEY METABOLISM ASSESSED IN INTACT KIDNEYS

Tissue metabolic activities are highly sensitive to the conditions of the tissue such as blood flow, oxygen supply, and tissue physiologic activities. Therefore assessment of metabolism in intact kidneys in vivo is essential for understanding kidney metabolism.

WHOLE KIDNEY BALANCE OF METABOLITES

Sampling of arterial blood, renal venous blood, and urine, coupled with unbiased metabolomic analysis, is a powerful approach for understanding the whole kidney handling of metabolites. The difference in the level of a metabolite in the renal venous blood compared with arterial blood is equal to the kidney production minus consumption and excretion of the metabolite.

In overnight-fasted and briefly anesthetized adult pigs (~50 kg), quantitative analysis of several hundred metabolites in arterial and renal venous serum and urine reveals that a large number of metabolites are cleared by the kidneys.[8] However, except for typical metabolic waste products such as uric acid and creatinine, a strong correlation was observed between the reabsorption efficiency of water-soluble metabolites and these metabolites' concentrations in the blood. In other words, more abundant metabolites are more effectively reabsorbed, which may contribute to preserving circulating metabolite levels. In these overnight-fasted and briefly anesthetized pigs, kidneys have a net production of glucose, likely because of the gluconeogenic capability of kidneys. Kidneys also produce glycocyamine, guanosine, allantoate, and phenylacetate. Kidneys release significant amounts of several amino acids.

Kidneys take up a significant amount of the TCA cycle intermediate citrate.[8] The amount of citrate taken up by kidneys is less only than lactate, uric acid, and glutamine. Tracing the fate of intravenously infused ^{13}C-labeled substrates reveals that citrate contributes up to 20% of the kidney TCA cycle, making citrate one of the largest contributors to the kidney TCA cycle. This is unique to kidneys as citrate does not contribute significantly to the TCA cycle in other organs. Other intravenously infused TCA cycle intermediates such as malate and succinate do not contribute significantly to the kidney TCA cycle.

Kidneys also consume medium and long-chain acylcarnitine and exhibit significant uptake of C5:0, C6:0, C8:0, C10:0, C10:1, C12:0, C12:1, C14:0, C14:1, C14:2, and C16:1 without release into urine.[8] In addition, kidneys use the short-chain fatty acid acetate, which may be produced by the gut microbiota and other organs, as fuel for TCA metabolism and a substrate for ketone body synthesis.

In a study of conscious, freely moving Sprague-Dawley rats, Shimada and colleagues[9] analyzed the metabolome in the arterial and renal venous blood and urine and calculated metabolite balance to identify metabolites that had net production or net consumption or loss by the kidney. They found that glucose was being net produced in the kidney in rats fed a normal-salt diet but not after the rats had been switched to a high-salt diet. The net production of lactate became significant at days 14 and 21 of the high-salt diet. A net consumption of citrate was found with the low-salt diet, which was sustained at most time points analyzed on the high-salt diet. There was a net consumption or loss of α-ketoglutarate in the low-salt–fed rats.

Few studies have simultaneously analyzed the metabolomes in arterial and renal venous blood in humans. Rhee and colleagues[10] collected arterial and renal venous plasma from nine individuals. The patients were referred for heart catheterization as part of their routine clinical care and were fasting at the time of their procedure. Plasma samples were obtained from a renal vein and the abdominal aorta before coronary artery catheterization. More than 200 metabolites were identified in these samples. Creatinine levels showed a renal venous-to-arterial (rV/A) ratio of 0.84. Several dozen metabolites showed rV/A ratios less than 1 consistently across subjects, indicating net loss of these metabolites as they pass through the kidney. Among them, hippuric acid, malate, xanthosine, and β-aminoisobutyric acid had rV/A ratios <0.50. Arginine (rV/A ratio 1.20) and serine (rV/A ratio 1.53) were among the metabolites that showed rV/A ratios >1 consistently across subjects, indicating net release of these metabolites from the kidney into the renal vein. Lipid metabolite levels were largely unchanged as they passed from the artery to the renal vein.

METABOLIC PATHWAY ACTIVITIES ASSESSED BY FLUXOMICS AND SPATIAL METABOLOMICS

The metabolism of a specific substance and the activities of the metabolic pathways involved can be assessed by tracing the metabolic products of the substance. The tracing can be achieved by approaches of fluxomics. In vivo fluxomics involves injecting a substance with stable isotope labeling into an organism, followed by analyzing the levels of isotope-labeled metabolites in the tissue of interest.

Hui and colleagues[11,12] performed fluxomic studies in mice for 15 ^{13}C-labeled circulating metabolites or groups of metabolites. Each of the isotope-labeled metabolites was infused at a rate that resulted in 10% to 15% final enrichment of tracer in serum. They measured tissue succinate and malate labeling to assess the direct contribution of each of the circulating metabolites to the TCA cycle in a given tissue, taking into account the circulating level of the metabolite and interconversion of circulating nutrients.

In the kidneys of fed mice on a carbohydrate diet, lactate accounts for 44% of the TCA cycle labeling, followed by approximately 20% from fatty acids and 10% each from glutamine and citrate. Short-chain fatty acid acetate, ketone body 3-hydroxybutyrate (3-HB), and a group of nonglutamine amino acids each contributes about 5%. Glycerol and glucose make small but appreciable contributions. The distribution of sources of tissue TCA metabolites in fasted mice on a carbohydrate diet is similar with fed mice, except for modest shifts to greater contributions of fatty acids, acetate, and 3-HB and fewer contributions of lactate and glucose. The glucose infusion resulted in strong labeling of lactate in the kidney, while the lactate infusion resulted in more labeling of TCA intermediates. This is consistent with the presence of glycolytic cells in the kidney that metabolize glucose to lactate and oxidative cells that use lactate to feed the TCA cycle.

It is important to note that the TCA contributions do not always correspond to tissue energy sources. In addition, fluxomic data are sensitive to experimental conditions including the amount of the tracer injected and the timing of tissue collection relative to the tracer injection. Finally, metabolic characteristics of mice may differ in some respects from larger mammals including humans. For example, mice devote a large fraction of energy to maintaining body heat.

Spatial metabolomics refers to the study of the metabolome in the spatial context of the tissue, which is enabled by advances in imaging mass spectrometry. Several lipids and other metabolites have been detected in kidneys using spatial metabolomic approaches. Spatial metabolomics may be combined with fluxomics to reveal metabolic pathway activities in specific regions or tissues in the kidney.

Wang and colleagues[13] infused ^{13}C- or ^{15}N-labeled nutrients into fasted mice and analyzed cryosections of the kidney using matrix-assisted laser desorption/ionization mass spectrometry imaging. Histologic or immunohistochemical analyses were performed in adjacent tissue sections to obtain anatomic information, and liquid chromatography (LC)–MS analysis was performed to validate mass spectrometry imaging signal assignments. Infusion of [U-^{13}C]glucose resulted in the production of ^{13}C6-UDP-glucose, an indication of glycolytic activities, primarily in the renal medulla. Infusion of [U-^{13}C]glycerol resulted in the production of ^{13}C3-UDP-glucose, an indication of gluconeogenesis, predominantly in the renal cortex. With malate as the readout, it was determined that lactate and citrate were the largest direct contributors to the TCA cycle carbon in the cortex. Lactate and free fatty acids were the largest direct contributors to the TCA cycle carbon in the medulla. In each kidney region, glucose contributes to TCA cycle intermediates primarily indirectly via circulating lactate. Glycerol and glutamine labeled the cortex more than the medulla. However, the detected labeling differences across kidney regions were modest for most substrates tested, which may be in part due to technical limitations.

The spatial labeling of the infused substrate and its downstream metabolites can be explained by the tissue distribution of transporters and enzymes in some cases.[13] For example, following the [U-^{13}C]glutamine infusion, glutamine itself was most intensely labeled in the medulla, while malate was most intensely labeled in the cortex. The medulla has low levels of protein expression of glutamine synthetase and glutaminase. Therefore the infused glutamine is not diluted or metabolized substantially in the medulla. The cortex has high levels of glutaminase and glutamate dehydrogenase expression, which explains the high degree of conversion of glutamine to downstream metabolites. These findings support the value of using gene expression or enzymatic activity analysis to predict in vivo metabolic activities in specific cell types or tissue regions in the kidney as detailed in later parts of this chapter.

Applying the combination of fluxomics and spatial metabolomics to study 350-μm-thick vibratome slices of fresh mouse kidney, Wang and colleagues[14] were able to measure several endogenous and ^{13}C-labeled metabolites from glycolysis and the TCA cycle, as well as branching metabolites. They found significant differences between the S1/S2 and S3 segments of the proximal tubule. The S3 segment, which is in kidney regions with lower oxygen levels, showed higher glycolytic activities and lower TCA cycle activities, compared to the S1/S2 segments. In addition, they found greater glycolytic activities and lower TCA cycle activities in proximal tubules in sections of kidneys from mice subjected to ischemia-reperfusion injury. Such metabolic disturbances were observed even in proximal tubules that appeared normal, which is reminiscent of the substantial molecular alterations observed in histologically normal glomeruli and tubulointerstitial regions in patients with focal segmental glomerulosclerosis.[15]

Spatial metabolomics, alone or in combination with fluxomics, holds great promise for revealing metabolic activities in specific cell types in vivo, which will be especially informative for histologically and metabolically complex organs such as the kidney. This approach will benefit from technologic advances that improve spatial resolution of the metabolite analysis, as well as detection and accurate quantification of a larger number of metabolites.

KIDNEY TISSUE OXYGENATION

Kidney metabolism is closely linked with kidney tissue oxygenation. Determinants of renal oxygenation and tissue oxygen tension (PO_2) include 1. RBF and oxygen content of arterial blood; 2. the oxygen consumed by the cells, which is largely determined by tubular transport activities; and 3. arterial-to-venous (AV) oxygen shunting, which entails the diffusion of oxygen from preglomerular arteries to postglomerular veins without being available to the cell for consumption.

The kidney receives a high blood flow, nearly 25% of the cardiac output, which is needed to sustain glomerular filtration rate (GFR). Compared with other major body organs, renal QO_2 (product of RBF and renal oxygen extraction) per gram of tissue is high, second only to the heart (2.7 mmol/kg/min vs. 4.3 mmol/kg/min for the heart).[16] The phenomenon of O_2 shunting from descending to ascending vasa recta in the medulla has been accepted,[17,18] yet little else is known regarding the location of shunting in the cortex or its impact on oxygenation.

Countercurrent flow in "hairpin loops" formed by the vasa recta facilitates the recycling of solutes to the inner medulla, where a high osmolarity is essential to the formation of concentrated urine (see Chapter 10). As an inherent consequence of this countercurrent mechanism for maintaining a medullary osmotic gradient, there arises a negative oxygen gradient from cortex to inner medulla, where PO_2 falls to 10 mm Hg.[19] This results from the combination of slow blood flow through the vasa recta, O_2 consumption by active transport in the outer medullary TAL, and diffusion of O_2 from descending to ascending vasa recta.[19] This leaves the medullary tissue at the brink of hypoxia, especially the stripe of outer medulla where the S3 segment of the proximal tubule and medullary TAL lie, making these segments most vulnerable to ischemic injury as they reabsorb significant fractions of filtered Na^+.

In most organs, tissue oxygen can be stabilized by metabolic regulation of blood flow. In such an arrangement, vasoactive end products of metabolism due to increased metabolic activity and oxygen utilization produce a signal that results in more blood flow to that organ. A unique feature of renal oxygenation is that the kidney cannot rely on this mode of metabolic autoregulation because, unlike other organs that receive blood solely to supply the metabolic needs of the organ, the kidney also receives blood to perform the functions of glomerular filtration and tubule transport.

RBF creates its own demand because it determines GFR, which in turn determines the rate of sodium reabsorption, which is the main determinant of QO_2.[20,21] If the kidneys were to modulate RBF as a means of stabilizing renal O_2 content, this would create a vicious cycle of positive feedback in which increased O_2 delivery would increase O_2 consumption, which would call for more O_2 delivery. Positive feedback is inherently destabilizing, so this arrangement alone could not work to stabilize either RBF or renal O_2 content. Hence the kidney is compelled to invoke mechanisms that are more complex. There are two generic routes for the kidney to stabilize its O_2 content. One is to dissociate RBF from GFR. The other is to alter the metabolic efficiency of Na^+ transport.

The rate at which the kidney consumes oxygen is linked to GFR. This is because the main use of oxygen is to support the reabsorption of the filtered sodium, which is linked to GFR by glomerulotubular balance (GTB). GTB describes the direct effect of the filtered load on tubular reabsorption, and it operates in all nephron segments, although the mechanism differs between segments. In the proximal tubule, shear strain tied to increased tubular flow exerts torque on the apical microvilli, which leads to upregulation of apical sodium transporters.[22,23] In cases in which filtration fraction increases, the parallel increase in peritubular capillary oncotic pressure will increase the Starling force driving fluid reabsorption. In the TAL, flux through NKCC2 is limited by chloride concentration, which declines more slowly along the TAL at high flow rates. On the other hand, increased flow rate in the tubule also shortens the time that a given sodium ion is exposed to the reabsorptive machinery. This leads to the prediction that GTB will just maintain constant fractional reabsorption.[21]

Renal regional tissue oxygenation levels in humans may be measured by blood oxygenation level–dependent magnetic resonance imaging (BOLD-MRI).[24] A BOLD-MRI analysis in 10 male normotensive humans and 8 untreated hypertensive patients found that a low-salt diet increased renal medullary tissue oxygenation levels in both groups when compared with a high-salt diet.[25] Renal cortical tissue oxygenation did not change in either group. In the normotensive group only, renal medullary oxygenation correlated positively with proximal tubular reabsorption of sodium estimated from measurements of the fractional excretion of lithium and negatively with distal sodium reabsorption. These data indicate that low sodium intake increases renal medullary oxygenation, which might involve greater proximal reabsorption and less reabsorption and possibly lower metabolic activities and oxygen consumption in the distal nephron in some humans.

Renal medullary tissue oxygenation levels were significantly lower in a group of 20 African Americans compared with 29 Whites.[26] The participants of this study had hypertension and were treated with an angiotensin-converting enzyme inhibitor or angiotensin receptor blocker. Daily sodium intake and urinary sodium excretion were the same between the two groups, while RBF, GFR, and the filtered sodium load were higher in the African American group, which would suggest greater tubular reabsorption of sodium in the African Americans. Renal cortical tissue oxygenation was similar between the two groups, but renal medullary tissue oxygenation levels were lower in the African Americans. Furosemide, which inhibits sodium reabsorption in the thick ascending limb, increased medullary tissue oxygenation to similar levels in the two groups, suggesting that the thick ascending limb in African Americans might have greater reabsorptive activities and consume more oxygen than in Whites. Blood pressure in African Americans is more likely to be salt sensitive than in Whites.

KIDNEY METABOLISM ASSESSED IN ISOLATED TISSUES OR USING INDIRECT METHODS

Studies of intact kidneys in vivo provide highly valuable information about kidney metabolism under physiologic conditions. However, the kidney is a histologically highly heterogeneous organ, and the nephron consists of several segments with distinct physiologic functions. The current technology for analyzing kidney metabolism in vivo does not allow comprehensive assessment of metabolism in each cell type in the kidney. Several studies examined metabolic pathway activities and the abundance of gene transcripts or proteins relevant to metabolism or enzyme activities in isolated nephron segments or various cell types in the kidney. These studies provide valuable insights into nephron segment– or cell type–specific metabolism.

Uchida and Endou[27] examined a range of substrates for their ability to maintain cellular ATP levels in microdissected glomeruli and nephron segments (excluding thin sections of loop of Henle and papillary duct). The substrates studied (all at 2 mmol/L) included L-glutamine, D-glucose, β-hydroxybutyrate, and DL-lactate. The change in ATP per millimeter of tubule (or glomerulus) as a function of substrate addition, shown in Fig. 5.5, illustrates that each segment had a distinct ability to use these substrates. Lactate was effective at maintaining ATP levels in all nephron segments tested, notably in the proximal tubule. The S1, S2, and S3 segments of the proximal tubule all used glutamine effectively as fuel, which is consistent with the role of the proximal tubule in ammoniagenesis. Glutamine is the main amino acid oxidized by the proximal tubule, where it is deaminated and converted to α-ketoglutarate, yielding 2 NH3 molecules that are secreted during acidosis, as illustrated in Fig. 5.4 and discussed in Chapter 9. Glutamine is not a preferred fuel in the more distal nephron segments. Glucose is completely reabsorbed along the proximal tubule, but glucose is not an effective metabolic fuel for the S1 or S2 regions of the proximal tubule. In contrast, all of the more distal segments tested readily used glucose to maintain cellular ATP. The ketone β-hydroxybutyrate was used effectively in all nephron segments tested; however, in S1 and S2 of the proximal tubule the capacity of β-hydroxybutyrate to support ATP production was far less than that provided by glutamine or lactate.

The distribution along the nephron of numerous enzymes involved in metabolic pathways, collated from many studies, has been summarized by Guder and Ross.[28] Their description of glycolytic (Fig. 5.6A) and gluconeogenic (see Fig. 5.6B) enzymes along the rat nephron[29-31] demonstrates low glycolytic potential in the proximal tubule and high glycolytic potential from medullary ascending limb to medullary collecting tubule. In contrast, gluconeogenic enzymes are found almost exclusively in the proximal tubule.

A summary of substrate preferences along the nephron is provided in Fig. 5.7[32] and discussed as follows.

PROXIMAL TUBULE

The proximal tubule reabsorbs approximately 65% of the filtered NaCl and water and nearly all filtered glucose and amino acids. Part of this reabsorption may occur passively through the paracellular space because of the presence of leaky tight junctions in the proximal tubule and the preferential reabsorption of bicarbonate in the early proximal tubule that leads to a chloride gradient favoring passive reabsorption. The reabsorption in the proximal tubule therefore may be energetically less expensive than other nephron segments that rely more on transcellular transport. Na^+-K^+-ATPase activity per unit length of the tubule segment is modest in the proximal tubule: lower than the thick ascending limb of the loop of Henle and the distal tubule but higher than other nephron segments (Fig. 5.8).

Mitochondrial volume relative to cytosolic volume and normalized surface area of mitochondrial inner membrane in the proximal tubule are similar to or slightly lower than the thick ascending limb of the loop of Henle and the distal tubule and higher than other nephron segments (see Fig. 5.8).

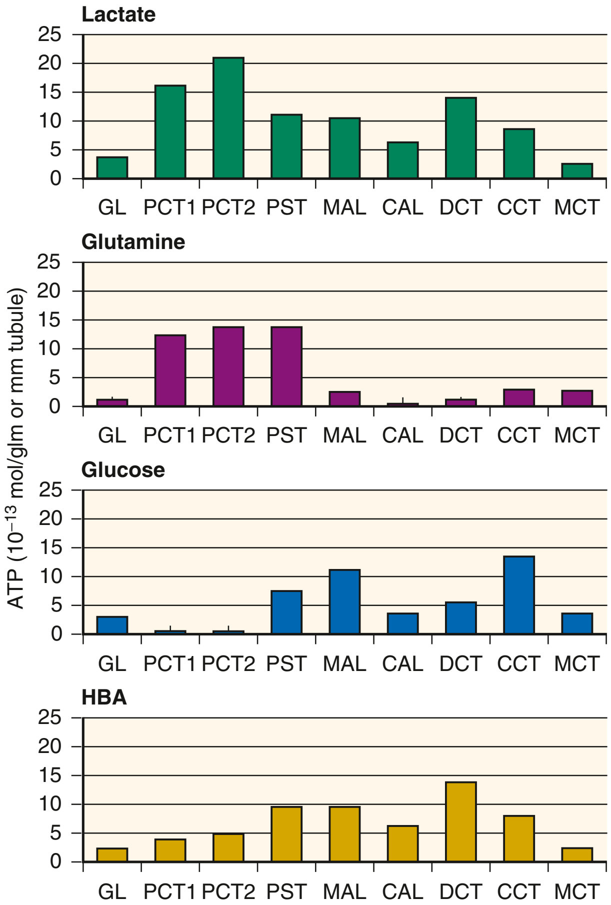

Fig. 5.5 Adenosine triphosphate (ATP) production in glomeruli and dissected nephron segments as a function of substrates. In glomeruli and PCT1, PCT2, and PST segments, the values equal the differences in ATP content between samples incubated with and without each substrate for 30 minutes. In MAL, CAL, DCT, CCT, and MCT, the values equal the differences in ATP content between samples incubated with and without each substrate in the presence of monensin (10 pg/mL) for 15 minutes. *CAL,* Cortical ascending limb; *CCT,* cortical collecting tubule; *DCT,* distal convoluted tubule; *GL,* glomerulus; *MAL,* medullary thick ascending limb; *MCT,* medullary collecting tubule; *PCT1,* early proximal convoluted tubule; *PCT2,* late proximal convoluted tubule; *PST,* proximal straight tubule. (Data from Uchida S, Endou H. Substrate specificity to maintain cellular ATP along the mouse nephron. *Am J Physiol*. 1988;255: F977–F983.)

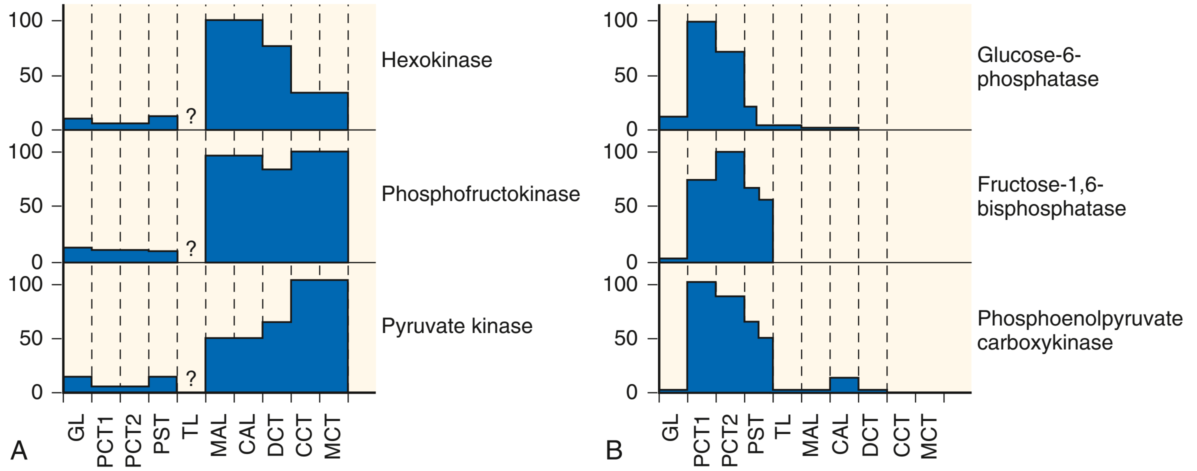

Fig. 5.6 Distribution of glycolytic and gluconeogenic enzymes along the rat nephron. Nephron segments were dissected from fed (A) and starved (B) rats, respectively. The activity of hexokinase, phosphofructokinase, pyruvate kinase, glucose-6-phosphatase, fructose 1,6-bisphosphatase, and phosphoenolpyruvate carboxykinase was determined in individual segments. Enzyme activities are expressed as a percentage of the maximal value observed, based on the original activity per gram of dry weight. *CAL,* Cortical ascending limb; *CCT,* cortical collecting tubule; *DCT,* distal convoluted tubule; *GL,* glomerulus; *MAL,* medullary thick ascending limb; *MCT,* medullary collecting tubule; *PCT1,* early proximal convoluted tubule; *PCT2,* late proximal convoluted tubule; *PST,* proximal straight tubule; *TL,* loop of Henle, thin limbs. (Modified from Guder WG, Ross BD. Enzyme distribution along the nephron. *Kidney Int*. 1984;26[2]:101–111.)

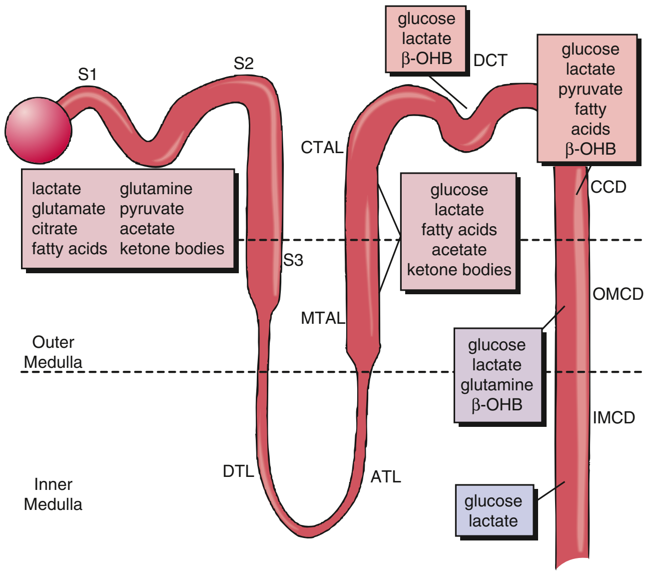

Fig. 5.7 Substrate preferences along the nephron. Summary of preferred substrates to fuel active transport in nephron segments as gleaned primarily from studies using oxygen consumption (Qo_2), ion fluxes, radioactive carbon (^{14}C)-labeled carbon dioxide generation from ^{14}C-labeled substrates, ATP contents, and reduced nicotinamide adenine dinucleotide fluorescence. *ATL,* Ascending thin limb; *β-OHB,* β-hydroxybutyrate; *CCD,* cortical collecting duct; *CTAL,* cortical thick ascending limb of the loop of Henle; *DCT,* distal convoluted tubule; *DTL,* descending thin limb; *IMCD,* inner medullary collecting duct; *MTAL,* medullary thick ascending limb of the loop of Henle; *OMCD,* outer medullary collecting duct; *S1, S2, S3,* successive segments of proximal tubule. (From Kone BC. Metabolic basis of solute transport. In *Brenner and Rector's the Kidney*. 5th ed. St Louis: Saunders; 2008.)

Distribution of mitochondrial enzymes, normalized to dry tissue weight or total protein, follows a similar pattern.

Free fatty acids appear to be a significant fuel for the proximal tubule. Other substances that the proximal tubule may use as fuel include glutamine, lactate, and ketone bodies. Glutamine catabolism in the proximal tubule generates ammonium ions, which plays a vital role in the renal excretion of excess acids. Glucose does not appear to be a major fuel in the proximal tubule. Instead, the proximal tubule has a significant gluconeogenetic capability, which may generate glucose from lactate and other substances including metabolites of amino acids and free fatty acids. The gluconeogenetic capability is unique to the proximal tubule and mostly absent in other nephron segments.

Studies carried out in a number of laboratories provide evidence that sodium transport and gluconeogenesis compete for ATP in the proximal convoluted tubule.[33-35] Friedrichs and Schoner[34] studied both processes in rat renal tubules and slices and found that ouabain inhibition of Na^+-K^+-ATPase increased renal gluconeogenesis by 10% to 40% depending on the substrate, and that stimulating Na^+-K^+-ATPase activity with high extracellular K^+ inhibited gluconeogenesis. The authors concluded that inhibition of the sodium pump induced a higher energy state of the cell, which would favor energy-requiring synthetic processes.

Nagami and Lee[35] used an isolated perfused mouse proximal tubule preparation to address this issue. When tubules were perfused at higher rates, delivering more sodium to the proximal tubule, the glucose production rate was decreased by 50%, whereas when tubules were incubated with ouabain in the bath or perfused with amiloride (to inhibit apical transport), the glucose production rate increased above that seen in nonperfused tubules. These authors also verified that the reduction in glucose production seen at elevated perfusion rates does not result from increased glucose utilization and is not dependent on the presence of specific substrates. Related to the topic of proximal tubule gluconeogenesis, two studies suggest that the blood glucose–lowering effects of sodium-glucose cotransporter isoform 2 (SGLT2) inhibitors are due, in part, to reduced proximal gluconeogenesis:

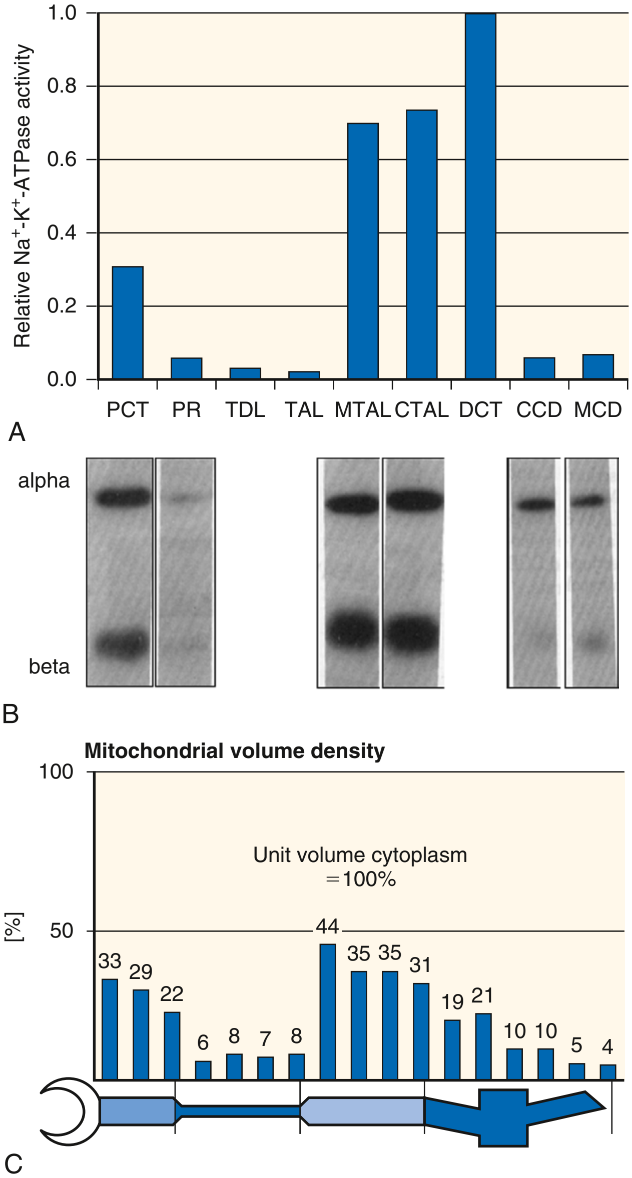

Fig. 5.8 Distribution of Na^+-K^+ATPase and mitochondrial density along the nephron. (A) Relative levels of Na^+-K^+ATPase activity measured in individual segments of the rat nephron. (Data are normalized to that of the distal convoluted tubule and expressed per unit length of tubule segment.) (B) Detection of Na^+-K^+-ATPase α_1- and β_1-subunits along the nephron. Tubule segments 40 mm long were resolved by sodium dodecyl sulfate–polyacrylamide gel electrophoresis and subjected to immunoblotting with subunit-specific antisera. Blots placed below corresponding tubule label indicated in (A). (C) Morphologic analysis of mitochondrial density relative to a unit of cytoplasm. *CCD,* Cortical collecting duct; *CTAL,* cortical thick ascending limb of the loop of Henle; *DCT,* distal convoluted tubule; *MCD,* outer medullary collecting duct; *MTAL,* medullary thick ascending limb of the loop of Henle; *PCT,* proximal convoluted tubule; *PR,* pars recta (proximal straight tubule); *TAL,* thin ascending limb of the loop of Henle; *TDL,* thin descending limb of the loop of Henle. ([A] redrawn from Katz AI, Doucet A, Morel F. Na+-K+-ATPase activity along the rabbit, rat, and mouse nephron. *Am J Physiol*. 1979;237:F114–F120; [B] based on data from McDonough AA, Magyar CE, Komatsu Y. Expression of Na[+]-K[+]-ATPase alpha- and beta-subunits along rat nephron: isoform specificity and response to hypokalemia. *Am J Physiol*. 1994;267:C901–C908; [C] based on data from Pfaller W, Rittinger M. Quantitative morphology of the rat kidney. *Int J Biochem*. 1980;12[1-2]:17–22.)

Inhibiting SGLT2-mediated glucose uptake in diabetic mice reduced gluconeogenic gene expression, including PEPCK, a principal regulator of gluconeogenesis.[36,37]

THICK ASCENDING LIMB

The thick ascending limb (TAL) of the loop of Henle reabsorbs approximately 20% to 25% of the filtered NaCl without reabsorbing water. The thin descending and ascending limbs of the loop of Henle do not have significant active transport. The TAL has a high rate of Na^+ transport against a steep concentration gradient, high levels of Na^+-K^+-ATPase activity and expression, and 40% of its cytosolic volume occupied by mitochondria (see Fig. 5.8). Although the TALs have a far greater capacity for anaerobic metabolism than the proximal tubules, this region still requires oxidative metabolism to maintain cellular ATP levels and active Na^+ reabsorption.[27,38] Glucose appears to be a significant fuel in the TAL. Lactate, fatty acids, and ketone bodies may also contribute. Glucose may be used to produce ATP via both glycolysis and oxidative phosphorylation in the TAL. Glycolytic capabilities are present in the TAL and subsequent nephron segments and largely absent in the proximal tubule.

DISTAL TUBULE AND COLLECTING DUCT

The distal tubule and the collecting duct reabsorb 5% to 10% of the filtered sodium and are the final segments that may control sodium excretion and urine flow rate. Substrate utilization in the cortical collecting duct is qualitatively similar to the TAL. The importance of glucose as the fuel appears to increase, and that of fatty acids decreases, as the collecting duct progresses to the renal inner medulla region.

CCD metabolism is particularly interesting because it is made up of distinctly different cell types: principal cells that reabsorb sodium and intercalated cells that can secrete bicarbonate (HCO_3^-). Hering-Smith and Hamm microperfused rabbit CCD and measured Na^+ reabsorption (with $^{22}Na^+$) from lumen to bath and HCO_3^- transport by microcalorimetry in the presence of substrates with or without inhibitors.[39] Both Na^+ reabsorption and HCO_3^- secretion were inhibited by antimycin A, which provides evidence for dependence on oxidative phosphorylation. However, neither was dependent on either glycolysis or the hexose–monophosphate shunt pathways. A small component of Na^+ transport was supported by endogenous substrates. Na^+ reabsorption was supported best by a mixture of basolateral glucose and acetate, whereas HCO_3^- secretion was fully supported by either glucose or acetate. HCO_3^- secretion (but not Na^+ transport) was supported to some extent by luminal glucose. In sum, this study indicates that principal cells and intercalated cells may have distinct metabolic phenotypes.

Medullary collecting ducts contribute to final urinary acidification. Comparing the outer medullary collecting duct (OMCD) with the CCD, Hering-Smith and Hamm[39] found that bicarbonate secretion in the OMCD could be fully supported by endogenous substrates. This region has far less sodium transport and few mitochondria (see Fig. 5.8). Stokes and colleagues[40] isolated IMCDs and examined their metabolic characteristics. In the absence of exogenous substrate, IMCD can maintain cellular ATP and respire normally, which is evidence for the presence of significant endogenous substrate. In the presence of rotenone, an inhibitor of oxidative phosphorylation, glycolysis increased 56%, which provides evidence for anaerobic metabolism, as supported by enzymatic profiles. Inhibition of sodium pump activity reduced QO_2 by 25% to 35%, which provides evidence for a linkage between sodium pump activity and oxidative metabolism.

In studies that examined the metabolic determinants of K^+ transport in isolated IMCD,[41] glucose increased both oxygen consumption and cell K^+ content by more than 10%, whereas an inhibitor of glycolysis promoted a release of cell K^+. Nor could cell K^+ content be maintained during inhibition of mitochondrial oxidative phosphorylation. Thus in the IMCD, both glycolysis and oxidative phosphorylation are required to maintain optimal Na^+-K^+-ATPase activity to preserve cellular K^+ gradients. Given the low PO_2 and low density of mitochondria in this region, the collecting ducts have a higher reliance on anaerobic metabolism but still take advantage of oxidative metabolism to fully support transport.

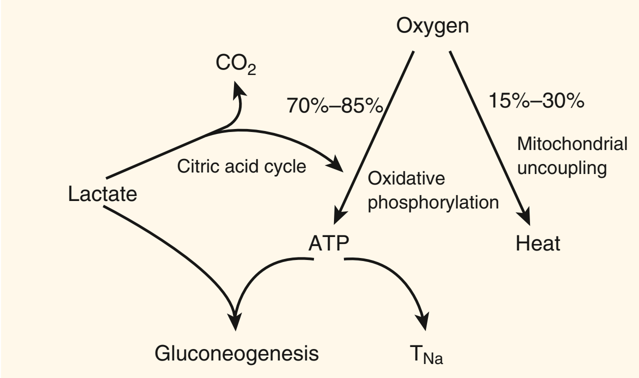

Fig. 5.9 Fate of lactate and oxygen in renal metabolism. Oxygen can be used to generate ATP via oxidative phosphorylation or heat if mitochondrial uncoupling occurs. Lactate can act as a substrate for gluconeogenesis, which consumes energy, or can enter into the citric acid cycle to generate energy. ATP is either used for Na transport (T_{Na}) or consumed in the process of gluconeogenesis.

PODOCYTE

Podocytes are an important part of the glomerular filtration barrier. In vivo podocytes have significantly lower mitochondrial abundance and decreased levels of TCA intermediates compared with tubular cells. In cultured podocytes, inhibition of the anaerobic glycolysis by administration of 2-DG or oxamate resulted in a significant decrease in the cellular ATP content, while inhibition of O_2-dependent pathways including fatty acid oxidation did not. These data suggest that podocytes may depend on the anaerobic glycolysis to meet their energy needs.[42] However, the role of oxidative phosphorylation and mitochondria may become prominent in podocytes in certain cell states or under disease conditions.[43]

IMPORTANCE OF RENAL GLUCONEOGENESIS

In a review of renal gluconeogenesis, Gerich and colleagues[44] comment that the kidney can be considered two separate organs, because the proximal tubule makes and releases glucose from noncarbohydrate precursors, whereas glucose utilization occurs primarily in the medulla. Because the kidney is both a consumer and a producer of glucose and glucose consumption in the medulla can mask glucose release by the cortex, net arteriovenous glucose differences across the kidney can be uninformative.

Gerich and colleagues[44] also make the case that the kidney is a significant gluconeogenic organ in normal humans based on the following: 1. In humans fasted overnight, proximal tubule gluconeogenesis can be as much as 40% of whole-body gluconeogenesis;[44] 2. during liver transplantation, endogenous glucose release falls to only 50% of control levels by 1 hour after liver removal;[45] and 3. pathologically in type 2 diabetes, renal glucose release is increased by about the same fraction as hepatic glucose release.[46]

RENAL HANDLING OF LACTATE

Lactate can reach the nephron by filtration or blood flow and can also be produced along the nephron. Within the kidney, lactate can be 1. oxidized to produce energy with generation of CO_2, a process that consumes oxygen but generates ATP; or 2. converted to glucose via gluconeogenesis in the proximal tubule, a process that consumes oxygen and ATP. This is shown in Fig. 5.9. Studies by Cohen[47] in an isolated whole kidney perfused with just lactate as substrate demonstrated a change in ^{14}C–lactate utilization as a function of its concentration in the perfusate: At low concentrations, all the lactate was oxidized (detected as CO_2) in order to fuel transport and basal metabolism; when lactate in perfusate was raised above 2 mmol/L, some of the lactate was used for synthesis of glucose (gluconeogenesis); and at high lactate in perfusate, the metabolic and synthetic rates approach maximum and some lactate is conserved (reabsorbed). However, it is not the normal circumstance that lactate is the sole substrate, and it is now appreciated that the metabolism of lactate is affected by the presence of other substrates—for example, lactate uptake and oxidation are inhibited in the presence of fatty acids.[6,48]

The kidney's ability to convert lactate to glucose provides evidence that it can participate in cell–cell lactate shuttle, also known as the Cori cycle.[49] This cycle is important when oxidative phosphorylation is inhibited in vigorously exercising muscle, which becomes hypoxic. In the muscle, pyruvate is reduced to lactate to regenerate NAD+ from NADH, which is necessary for ATP production by glycolysis to continue. Lactate is released into the blood and can be taken up by tissues capable of gluconeogenesis, such as the liver and kidney. In the proximal tubule, the lactate that is not oxidized can be converted to glucose, and because this substrate is not used by the proximal tubule, glucose will be reabsorbed back into the blood, where it will be available for metabolism by the exercising muscle. Overall, this cycle is metabolically costly: Glycolysis produces 2 ATP molecules at a cost of 6 ATP molecules consumed in the gluconeogenesis. Thus the Cori cycle is an energy-requiring process that shifts the metabolic burden away from the exercising muscle during hypoxia. This cell–cell lactate shuttle could also operate within the kidney between nephron segments that produce lactate anaerobically and the proximal tubule.

Renal medullary lactate concentration was explored in a 1965 study in rats by Scaglione and colleagues[50] to test the idea that the medulla used glycolysis in the low-oxygen environment. Medullary lactate concentration is a function of delivery via the blood flow, production in the medulla, and removal by the blood flow because there is no gluconeogenesis in this region to consume lactate. Because of the countercurrent arrangement of the vasa recta, lactate would be expected to concentrate in the medulla somewhat. The study results indicated that lactate concentration was twice as high in the inner medulla as in the cortex and that during osmotic diuresis the medullary lactate doubled, whereas cortical lactate remained unchanged. The authors postulated that increased medullary lactate was evidence for increased glycolysis during osmotic diuresis because the diuresis and increased flow through the vasa recta would be expected to decrease medullary lactate if synthesis rates were unchanged. Sodium delivery to the distal nephron would also increase during osmotic diuresis, and the accompanying increased Na^+ reabsorption could drive the increased glycolysis.

Bagnasco and colleagues[51] studied lactate production along the nephron in dissected rat nephron segments incubated in vitro with glucose with or without an inhibitor of oxidative metabolism, antimycin A. The only pathway for lactate production in the kidney is from pyruvate via lactate dehydrogenase. Proximal tubules produced no lactate with or without antimycin A. The distal segments all produced lactate, and the production was significantly increased (approximately 10-fold in TAL) during antimycin A incubation (Fig. 5.10), which led the authors to conclude that significant amounts of lactate can be produced by anaerobic glycolysis during anoxia in the distal segments. The IMCD, a region with low oxygen tension under control conditions, had high levels of lactate production even without antimycin A, which indicates that it is primed for anaerobic glycolysis.

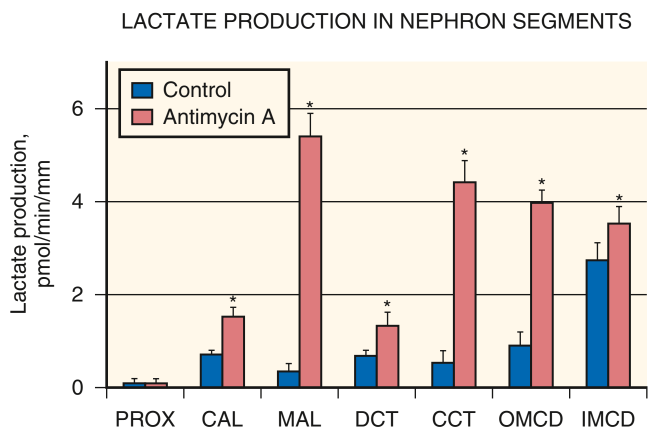

Fig. 5.10 Lactate production by rat nephron segments under control conditions and in the presence of antimycin A. *CAL,* Cortical ascending limb; *CCT,* cortical collecting tubule; *DCT,* distal convoluted tubule; *IMCD,* inner medullary collecting duct; *MAL,* medullary thick ascending limb; *OMCD,* outer medullary collecting duct; *PROX,* proximal tubule. (From Bagnasco S, Good D, Balaban R, et al. Lactate production in isolated segments of the rat nephron. *Am J Physiol*. 1985;248:F522–F526.)

INSIGHTS FROM GLOBAL EXPRESSION ANALYSIS OF ISOLATED NEPHRON SEGMENTS

Results from global analysis of isolated nephron segments using techniques like RNA-seq or proteomics confirmed many of the findings from previous, targeted analyses of metabolic activity and enzyme abundance and provided additional insights into metabolism in nephron segments. An RNA-seq analysis of 14 renal tubule segments isolated from rat kidneys confirmed that the proximal tubule lacked key glycolytic enzymes (hexokinase, phosphofructokinases, and pyruvate kinase) and strongly expressed mRNAs for enzymes that are critical for gluconeogenesis.[52] The study indicates that hexokinase 1 and phosphofructokinases are expressed in all segments beyond the proximal tubule, including the thin limbs of the loop of Henle, suggesting that glycolysis may be important for ATP production in the thin limbs. The proximal tubule also strongly expresses the key enzymes for fructolysis (fructokinase and dihydroxyacetone kinase 2) and is the main segment that expresses argininosuccinate synthase that is necessary for arginine synthesis. The kidney is a major producer of arginine in the body. The TAL and distal convoluted tubule (DCT) show particularly high mRNA abundance of carnitine palmitoyl transferases, key enzymes for fatty acid metabolism.

A proteomic analysis, again, confirmed many of the known metabolic characteristics of rat nephron segments.[53] The study shows selective expression of the osmoprotective enzyme aldose reductase in inner medullary segments and rate-limiting enzymes for ammoniagenesis in proximal tubule cells and maximum abundance of creatine kinase in segments with the highest rates of sodium reabsorption (mTAL, cTAL, DCT) or transport against a large Na gradient (CNT, CCD).

KIDNEY METABOLISM AND THE BIOENERGETICS OF TUBULAR TRANSPORT

Most of the energy provided by renal metabolism is committed to epithelial transport, which determines the volume and composition of the urine. Energy is required to form urine that differs in solute composition from that of the plasma.

Na^+-K^+-ATPase

Na^+-K^+-ATPase, also referred to as the "sodium pump," is a ubiquitous plasma membrane protein that transports intracellular sodium out of the cell and extracellular potassium into the cell, thereby generating opposite concentration gradients for sodium and potassium ions across the cell membrane. This process of separating sodium from potassium across the cell membrane is fueled by the hydrolysis of ATP.[54,55] Each cycle of the pump consumes 1 ATP molecule while transporting 3 Na^+ and 2 K^+ ions across the cell membrane. The hydrolysis of ATP and the associated transport of ions are mutually dependent[54,55] and constitute an example of primary active transport. In this process, there is nearly full conversion from chemical to mechanical energy, with minimal dissipation. The translational energy that develops after ATP hydrolysis results from electrostatic repulsion between the product ions, ADP and Pi, in accordance with Coulomb's law. Although this energy could be dissipated through subsequent collisions, such events are unlikely over short time scales and short distances. For a relative kinetic energy of the phosphate of 0.6 eV, for example, the phosphate ion moves about 0.1 nm in 0.3 ps. If no other collisions occur in that short time interval, the phosphate can then transfer its entire kinetic energy to the sodium pump in the form of a molecular strain. Given the intrinsic free energy of ATP hydrolysis, the pump can generate gradients that store up to approximately 0.6 eV of electrochemical potential per 3 Na^+ plus 2 K^+ ions. For a typical cell in a typical environment, about 0.4 eV is required to cycle the pump against the existing Na and K gradients, which means that cells tend to operate with some reserve to further reduce their sodium or increase their potassium concentrations.

The sodium pump is composed of an α catalytic subunit, which hydrolyzes ATP and transports Na^+ and K^+ across the membrane, a β-subunit that is critical for functional maturation and delivery of Na^+-K^+-ATPase to the plasma membrane, and an FXYD protein that can modulate the kinetics of Na^+-K^+-ATPase in a tissue-specific manner.[56] There are multiple isoforms of each subunit. The α1β1 heterodimer is likely the exclusive Na^+-K^+-ATPase in renal epithelia,[57] whereas several FXYD protein subunits are expressed differentially along the nephron.[56-59] Biophysical models describing the turnover of the sodium pump through its functional cycle are described in a review by Horisberger.[54]

Besides Na^+-K^+-ATPase, additional ion-translocating ATPases are expressed in renal epithelia along the nephron, including H^+-K^+-ATPase, Ca^{2+}-ATPases, and H^+-ATPases. These transport ATPases play important roles in maintaining urinary acidification and calcium homeostasis as discussed in Chapters 6, 7, and 9. These ATPases do not contribute significantly to the reabsorption of the bulk of the filtrate.

HARNESSING THE SODIUM POTENTIAL FOR WORK

For a cell in a steady state, the pumping of ions by the Na^+-K^+-ATPase must be offset by an equal and opposite diffusion of those ions back across the cell membrane. Because cell membranes are generally more permeable to potassium than sodium, potassium diffusion contributes more to the cell voltage than sodium diffusion, even though three sodium ions leak into the cell for every two potassium ions that leak out. Thus diffusion of potassium out of the cell dominates the cell voltage, making it negative. The net outcome of this pump-leak process is that electrochemical potential, which originates with ATP hydrolysis, becomes concentrated in the transmembrane sodium gradient, whereas potassium resides near electrochemical equilibrium.

The difference in electrochemical potential for sodium across the cell membrane is available to drive the unfavorable passage of other solutes across the membrane by a variety of exchangers and cotransporters. Examples include the proximal tubule Na^+/H^+ exchanger (NHE), sodium-glucose cotransporters (SGLTs), the basolateral Na/α-ketoglutarate (α-KG) cotransporter, the furosemide-sensitive Na-K-2Cl cotransporter (NKCC2), and the thiazide-sensitive Na-Cl cotransporter (NCC). Generically, transport that directly uses free energy from the sodium gradient to drive uphill flux of another solute is referred to as secondary active transport.[60] Tertiary active transport refers to the net flux of a solute against its electrochemical potential gradient coupled indirectly to the Na+ gradient (three transport processes working in parallel). An example of tertiary active transport is the uptake of various organic anions from the peritubular blood into the proximal tubular cell by the so-called organic anion transporters (OATs). Energy from the sodium gradient is converted into a gradient for α-KG to diffuse into the cell by Na/α-KG cotransport. OATs use this potential difference to exchange α-KG for another organic anion.[61]

For tubular cells that actively reabsorb chloride, free energy is transferred from the Na potential to drive apical chloride entry and raise cell chloride above equilibrium. In the proximal tubule, the energy for apical chloride entry is derived circuitously via sodium-hydrogen exchange that is coupled to oxalate, formate, or hydroxyl ion transport (see Chapter 6). In the thick ascending loop of Henle and DCT, the energy transfer occurs by direct cotransport with Na via NKCC2 or NCC. In each case, raising cell chloride above equilibrium provides a driving force for chloride to diffuse out of the cell across the basolateral membrane, which is permeable to chloride. Raising cell chloride also makes the basolateral membrane voltage less negative, as is apparent from the Goldman equation. Because luminal voltage is the sum of voltage steps across the basolateral and apical membranes, raising cell chloride in a cell with basolateral chloride conductance will raise the lumen voltage (make it more positive), thus providing free energy that can either be dissipated by the intercellular back-leak of chloride, which would increase entropy, or be applied to do the useful work of cation reabsorption, which would decrease entropy. The kidney uses the latter mechanism of energy transfer to augment Na reabsorption in the proximal tubule, as well as calcium and magnesium reabsorption in the thick ascending loop of Henle.

For cells that express epithelial Na channel (ENaC), opening these channels will depolarize the apical membrane, as can be seen from the Goldman equation. K ions, which enter the cell via the basolateral Na pump, can leave the cell by K conductance in either basolateral or apical membranes. Depolarizing the apical membrane will increase the fraction of K ions leaving by way of the apical membrane conductance. This represents the transfer of free energy from the Na^+-K^+-ATPase and the apical Na potential to the useful work of K secretion.

CELL POLARITY AND VECTORIAL TRANSPORT

The polar arrangement of transporters in renal cells is essential for vectorial transport. Wherever it is expressed along the nephron, the sodium pump, which removes sodium from the cell, is restricted to the basolateral membrane. Meanwhile, the variety of exchangers, cotransporters, and sodium channels through which sodium enters the tubular cell are restricted to the apical membrane. These include the principal Na+/H+ exchanger NHE3 and SGLTs in the proximal tubule, the NKCC2 in the TAL of the loop of Henle, the NCC in the DCT, and epithelial sodium channels in the connecting tubule and collecting duct (see Chapter 6). These apical sodium transporters affect secondary active transport coupled to the primary active transporter, Na^+-K^+-ATPase.

Close coordination of sodium uptake across the apical membrane with sodium extrusion across the basolateral membrane is required to avoid osmotic swelling and shrinking of the cell. Assuming ATP is not limiting for basolateral exit, the magnitude of transepithelial transport is a function of 1. the number of transporters in the plasma membrane, which can be varied by changes in synthesis or degradation rates and/or trafficking between intracellular and plasma membranes, and 2. the activity per transporter, which can be varied by covalent modification (e.g., phosphorylation or proteolysis) or protein–protein interaction (e.g., Na^+-K^+-ATPase kinetics are influenced by FXYD subunit association).[56] The rate of apical sodium entry is also subject to influence by the availability of substrates for cotransport. For example, the amount of sodium-glucose cotransport depends on the availability of glucose in proximal tubular fluid, and the sodium entry at a given point along the TAL is subject to variations in the local chloride concentration because NKCC2 has a relatively low affinity for chloride.

Many factors and hormones known to regulate renal sodium reabsorption (including angiotensin II, aldosterone, dopamine, parathyroid hormone, and blood pressure) act in parallel to affect the activity, distribution, or abundance of apical transporters and basolateral sodium pumps.[57,62] The molecular basis of this apical–basolateral crosstalk is not clearly understood, especially in the light of close cell volume control; however, there is evidence for the role of elevated cellular calcium level in response to depressed sodium transport.[63] There is also evidence for a salt-inducible kinase that responds to slight elevations in cell Na and Ca,[64] as well as evidence for coupling of Na^+-K^+-ATPase to apical channel activity.[65]

WHITTAM MODEL

In the early 1960s, the coupling between active transport, respiration, and Na^+-K^+-ATPase activity was recognized by Whittam and Blond,[66,67] who tested the idea that inhibition of active ion transport at the plasma membrane would cause a fall in oxygen consumption (QO_2) in the mitochondria. Using brain or kidney samples studied in vitro, they demonstrated that inhibition of Na^+-K^+-ATPase activity by removal of sodium or addition of the sodium pump–specific inhibitor ouabain (neither of which directly inhibits mitochondrial respiration) markedly reduced QO_2, which led the investigators to conclude that an extramitochondrial ATPase, sensitive to Na^+ and ouabain, as well as to K^+ and Ca^{2+}, is one of the pacemakers of respiration of the kidney cortex.[66,67]

A study by Balaban and colleagues 2 decades later[68] used a suspension of renal cortical tubules to reexamine this Whittam model (see Fig. 5.3) in more detail by measuring the redox state of mitochondrial nicotinamide adenine dinucleotide (NAD), cellular ATP and ADP concentrations, ATP/ADP ratio, and QO_2 in the same samples. If transport and respiration are assumed to be coupled, inhibition of transport is predicted to provoke a mitochondrial transition to a resting state[69] accompanied by an increase in NADH/NAD^+ (reduced to oxidized NAD), increase in [ATP], decrease in [ADP] and [Pi], increase in ATP/ADP ratio, and decrease in QO_2. Stimulation of active transport would provoke the opposite pattern: decreased NADH, ATP, and ATP/ADP ratio and increased QO_2. Predictably, incubating the renal cortical tubule suspension with the Na^+-K^+-ATPase inhibitor ouabain caused a 50% decline in QO_2, reduction of NAD to NADH, and a 30% increase in the ATP/ADP ratio, all evidence for coupling of mitochondrial ATP production to ATP consumption via Na^+-K^+-ATPase. Similarly, in tubules deprived of K^+ (which is required for Na^+-K^+-ATPase turnover), adding 5 mmol/L K^+ increased QO_2 by more than 50%, oxidized NADH to NAD^+, and decreased the cellular ATP/ADP ratio by 50%. These results provide evidence for the coupling of both Na^+-K^+-ATPase and ATP production via ATP synthase to the cellular ATP/ADP ratio (see Fig. 5.3).

ENERGY REQUIREMENTS ALONG THE NEPHRON

Despite consistent distribution and function, the relative abundance of Na^+-K^+-ATPase as a function of tubular location along the nephron is highly variable. Na^+-K^+-ATPase activity, ouabain binding, and Na^+-K^+-ATPase subunit abundance have been studied in dissected tubules and with imaging techniques. Na, K-ATPase expression patterns and ouabain binding patterns along the nephron are similar.[70,71] The pronounced differences in activity can largely be accounted for by differences in sodium pump number measured either by ouabain binding or by immunoblot of subunits in dissected nephron segments (see Fig. 5.8).[72]

The patterns of Na^+-K^+-ATPase protein expression and activity as a function of tubule length are what is to be expected from what is understood of the physiology of the nephron segments: Moderate levels are expressed in the proximal tubule, where two-thirds of the sodium is reabsorbed across a leaky epithelium, and lower levels are expressed in the straight than in the convoluted segments, reflecting the amount of sodium transported in these two regions. Low levels are detected in the thin limbs of the loop of Henle, whereas high levels are expressed in the medullary and cortical TAL that reabsorbs a significant fraction of NaCl without water against an increasingly steep transepithelial gradient. The Na^+-K^+-ATPase activity and expression in the DCT, which is responsible for reabsorbing another 5% to 7% of the filtered load against a steep transepithelial gradient, is high. In the collecting duct, which reabsorbs a smaller fraction of Na^+ via channels electrically coupled to the secretion of K^+ or H^+ and has variable H_2O permeability, the Na^+-K^+-ATPase is quite low, albeit sufficient to drive sodium reabsorption in this region. The distribution of the ATP-producing mitochondria along the nephron, reported as percent of cytoplasmic volume,[73] parallels the distribution of the ATP-consuming sodium pumps but is somewhat less variable, ranging from 10% or less of the cell volume in the thin loop of Henle and medullary collecting duct to 20% in the cortical collecting duct (CCD) and proximal straight tubule to 30% to 40% of cell volume in the proximal convoluted tubule and TAL[73] (see Fig. 5.8).

SEXUAL DIMORPHIC PATTERN OF TRANSPORTERS ALONG THE NEPHRON AND METABOLIC CONSIDERATIONS

For 30 years, sex differences in renal hemodynamics including lower GFR and higher RVR in females and similar blood pressures between sexes have been recognized

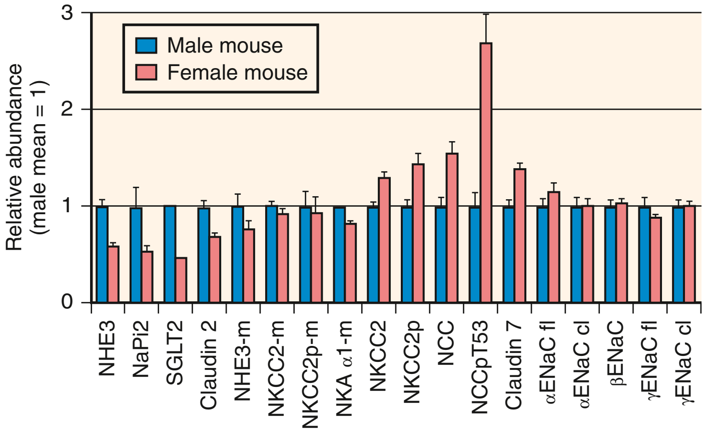

Fig. 5.11 Profile of renal sodium transporter protein abundance in female C57BL/6 mice expressed relative to abundance in male mice (defined as 1.0). Cortical and medullary (*-m*) samples were detected by immunoblot: Na^+/H^+ exchanger isoform 3 (*NHE3*), Na^+-phosphate cotransporter isoform 2 (NaPi2), Na^+-glucose cotransporter (*SGLT2*), claudin 2, Na^+-K^+-$2Cl^-$ cotransporter isoform 2 (*NKCC2* and activated phosphorylated form *NKCC2p*), Na,K-ATPase α catalytic isoform, Na^+-Cl^- cotransporter (*NCC* and activated phosphorylated form NCCp), claudin 7, epithelial Na^+ channel α,β,γ subunits full length *(fl),* and activated cleaved *(cl)* forms. (Adapted from Veiras LC, Girardi ACC, Curry J, et al. Sexual dimorphic pattern of renal transporters and electrolyte homeostasis. *J Am Soc Nephrol*. 2017;28:3504–3517 and Pastor-Soler NM, Hallows KR. AMP-activated protein kinase regulation of kidney tubular transport. *Curr Opin Nephrol Hypertens*. 2012;21:523–533.)

in experimental rodents.[74] Immunoblots coupled with physiologic assays in rats indicate that females, versus males, exhibit a distinct transporter profile of lower proximal transporters' abundance and HCO_3^- reabsorption coupled to higher distal NCC and ENaC activation[75] (Fig. 5.11). This shift in T_{Na} from the energetically efficient PT to the costlier distal nephron is predicted to decrease sodium transport energy efficiency (T_{Na}/Qo_2) in females. A rationale for this downstream shift in T_{Na} can be found in female biology. Pregnancy, and even more so lactation, represent major challenges to fluid homeostasis in females. The proximal tubule, which is shorter at baseline in females versus males,[76] lengthens during lactation, driving a proportional increase in T_{Na} in a region where transport efficiency is high due to paracellular T_{Na}.[76,77] These significant sex- and reproduction-dependent differences in renal transport function likely necessitate differences in nephron region–specific metabolism that warrant serious future consideration.

METABOLIC COST OF SODIUM REABSORPTION

The cost of renal sodium transport can be estimated from the sodium pump stoichiometry and amount of oxygen required to produce ATP. Sodium pump stoichiometry dictates that hydrolysis of one ATP molecule is coupled with the transport of 3 Na^+ ions out of the cell and 2 K^+ ions into the cell,[54] and oxidative metabolism generates approximately 6 ATP molecules per O_2 molecule consumed (see Table 5.1 and Fig. 5.4). In the 1960s, several investigators undertook measuring the metabolic cost of tubular reabsorption in various species of mammals. There is fair consensus among four oft-cited studies published between 1961 and 1966 that the relationship between Qo_2 and T_{Na} is linear and that the kidney reabsorbs 25 to 29 Na^+ ions per molecule of O_2 consumed in the process.[78-81]

If one assumes that kidney mitochondria make 6 molecules of ATP per molecule of O_2, the kidney must then reabsorb 4 to 5 Na^+ per ATP molecule. This exceeds the 3:1 stoichiometry of the Na^+-K^+-ATPase, which was known at the time (reviewed

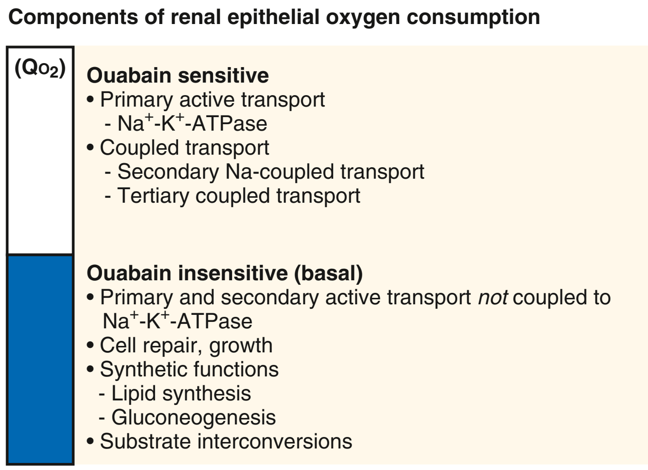

Fig. 5.12 Ouabain-sensitive and -sensitive oxygen consumption in the kidney. A large fraction of renal epithelial oxygen consumption (Qo_2) in renal cells is sensitive to the Na^+-K^+-ATPase–specific inhibitor ouabain, and this Qo_2 drives primary active transport and transport coupled with sodium pump activity. The fraction of renal oxygen consumption that does not change in the presence of ouabain is, by definition, independent of Na^+-K^+-ATPase activity in the cell and is roughly equivalent to the basal Qo_2, which fuels transport not coupled to sodium gradients, cell repair and growth, biosynthesis, and substrate interconversions.

in Burg and Good).[82] In fact, one might expect a ratio lower than 3 because of the metabolic needs of the kidney that are independent of sodium transport (i.e., insensitive to the Na^+-K^+-ATPase inhibitor ouabain) (Fig. 5.12) and because of tubular backleak. Because there are thermodynamic difficulties with the idea of an undiscovered basolateral sodium pump capable of forcing 5 Na^+ from a tubular cell with energy from a single ATP molecule, it was surmised that a considerable fraction of overall sodium reabsorption must be passive and paracellular, as is now accepted.

The kidney can leverage excess free energy in the gradients created by primary and secondary active transport to drive passive paracellular reabsorption of sodium chloride. The paracellular route in the proximal tubule responsible for sodium reabsorption is the tight junction protein channel claudin-2.[83] Free energy for paracellular reabsorption is available in the midproximal tubule and early TAL.[82] In the proximal tubule, the driving force for the passive transport develops as a result of the preferential absorption of bicarbonate over chloride earlier in the tubule.[84,85]

The decline in tubular bicarbonate concentration is paralleled by a rise in chloride concentration as water follows Na^+, HCO_3^-, and organic osmoles across the leaky proximal tubule (see Chapter 6). This favorable lumen-to-blood Cl^- gradient drives passive paracellular chloride reabsorption. The transepithelial voltage that arises from electrodiffusion of chloride, in turn, drives passive sodium reabsorption. Because the NaCl reflection coefficient is less than that for $NaHCO_3$ in this region,[84] coupled sodium chloride reabsorption also occurs secondary to solvent drag.[86] Although estimates vary, this passive reabsorption may increase the number of Na^+ ions reabsorbed to O_2 molecules consumed in the proximal tubule from 18 to 48.[87] Blunting paracellular reabsorption in the claudin-2–null mouse increases the oxygen consumption cost of T_{Na} by shifting it from paracellular to transcellular reabsorption, supporting the idea that paracellular transport is required for energy efficiency of oxygen utilization in the service of the large fraction of Na^+ reabsorption that occurs in the proximal tubule.[88]

The early portion of the TAL is also capable of paracellular Na^+ reabsorption. In this region, Na^+ can be transported transcellularly by the apical Na-K-2Cl cotransporter or apical Na^+/H^+ exchanger, secondary to a high density of Na^+-K^+-ATPase extruding Na^+ across the basolateral membranes. In addition, Na^+ can be reabsorbed paracellularly as long as there is a lumen-positive transepithelial voltage sufficient to overcome the force for back diffusion associated with an unfavorable concentration difference. A lumen-positive voltage develops in the TAL because the apical membrane has a high concentration of K^+ channels, whereas the basolateral membrane has both K^+ and Cl^- channels. As predicted by the Goldman-Hodgkin-Katz voltage equation, the chloride conductance causes the basolateral membrane potential to be less negative than the apical membrane potential, which results in a positive transepithelial gradient.[89,90]

Further along the nephron in the distal tubule and collecting duct, the tubular fluid sodium concentration is too low to allow paracellular reabsorption of sodium. In those segments, a lower limit on the cost of Na^+ reabsorption is set by the 3 Na^+/1 ATP ratio of the sodium pump.

Energetic efficiency of Na^+ reabsorption is highly variable along the nephron, in large part due to the presence of paracellular pathways for reabsorption in "leaky" epithelia, such as the proximal tubule, that can take advantage of favorable concentration and electrochemical gradients to reabsorb Na^+ without requiring ATP hydrolysis. Thus if proximal reabsorption is lower (as in females) or reduced, the shift of T_{Na} to more distal sites should provoke an increase in Qo_2 driven by higher active uphill transcellular Na^+ reabsorption. The example of the cldn-2 null mouse is provided earlier as an example.[88] Layton and colleagues[91,92] now provide computational models that assess how shifting T_{Na} from one region to another or inhibiting T_{Na} in a region-specific manner alters overall Qo_2. In this model, increasing single-nephron GFR did not alter the efficiency of reabsorbing Na^+ in the proximal tubule (T_{Na}/Qo_2) while it increased T_{Na}/Qo_2 downstream secondary to higher paracellular T_{Na} due to higher luminal [Na^+]. This finding suggests that the proximal S3 segment and medullary TAL (which are vulnerable to hypoxia) are buffered from higher Qo_2 in response to increased flow and substrate delivery.[91] When the model was applied to examine the effects of inhibiting individual transporters, the impact of inhibiting the proximal tubule NHE3 was most pronounced, not unexpected given the fact that the NHE3 reabsorbs 36% of filtered Na^+ at baseline. Overall, inhibiting NHE3 by 80% reduced GFR by 30% (mediated by tubuloglomerular feedback) and reduced whole kidney T_{Na}/Qo_2 by 20%, explained by far less proximal paracellular T_{Na} and the shift of T_{Na} to less efficient distal regions. Interestingly, the model predicts that NHE3 and NKCC2 inhibition increase CNT and CD ENaC-mediated T_{Na}/Qo_2 secondary to increased luminal Na^+. Inhibition of NKCC2, NCC, or ENaC, although changing urine volume, Na^+ and K^+ excretion as predicted, did not change whole kidney T_{Na}/Qo_2.[92]

Although active transport of sodium is a pacemaker for renal respiration, there are ways to reset the relationship of Qo_2 to sodium pump activity. Examples of this were provided by Silva and Epstein, who measured both O_2 consumption and Na^+-K^+-ATPase activity in rat kidney slices in which an increase in the latter had been induced by prior treatment of the animals with triiodothyronine (T_3), methylprednisolone, potassium loading, or subtotal nephrectomy. Although each of these maneuvers increased ex vivo sodium pump activity, only T_3 and methylprednisolone increased Qo_2.[93]

It has also been shown that the thermogenic effect of catecholamines, normally associated with brown fat and striated muscle, also occurs in the kidney, which responds to dopamine infusion with a near doubling of overall metabolic rate but minimal change in sodium reabsorption.[94] Dopamine inhibits Na^+ reabsorption in the proximal tubule,[95,96] thereby shifting the reabsorptive burden to less efficient downstream segments. However, heat accumulates in both cortex and medulla during dopamine infusion, which suggests that the mechanism may be a direct effect of catecholamines on renal metabolism.

Weinstein and Szyjewicz[97,98] also examined Qo_2/T_{Na} using 10% body weight short-term saline expansion as another way to inhibit proximal Na^+ reabsorption in rats. This maneuver reduced fractional Na^+ reabsorption by 30% in the proximal tubule, leading to an increase in net reabsorption downstream of the proximal tubule. Yet overall Qo_2 did not increase; it actually fell. It was conjectured that energy for this increase in downstream reabsorption was derived anaerobically, but the full details of this remain to be clarified. It appears that the energy cost of transport in the proximal versus distal nephron during inhibition of proximal tubule transport depends on the nature of the stimulus provoking the change in transport, as well as the metabolic environment.

FILTRATION FRACTION AND OTHER REGULATORS OF KIDNEY OXYGENATION AND QO_2/T_{NA}

As discussed earlier, there are two generic routes for the kidney to achieve a stable content of O_2: dissociation of RBF from GFR and alteration of Qo_2/T_{Na}. Both routes

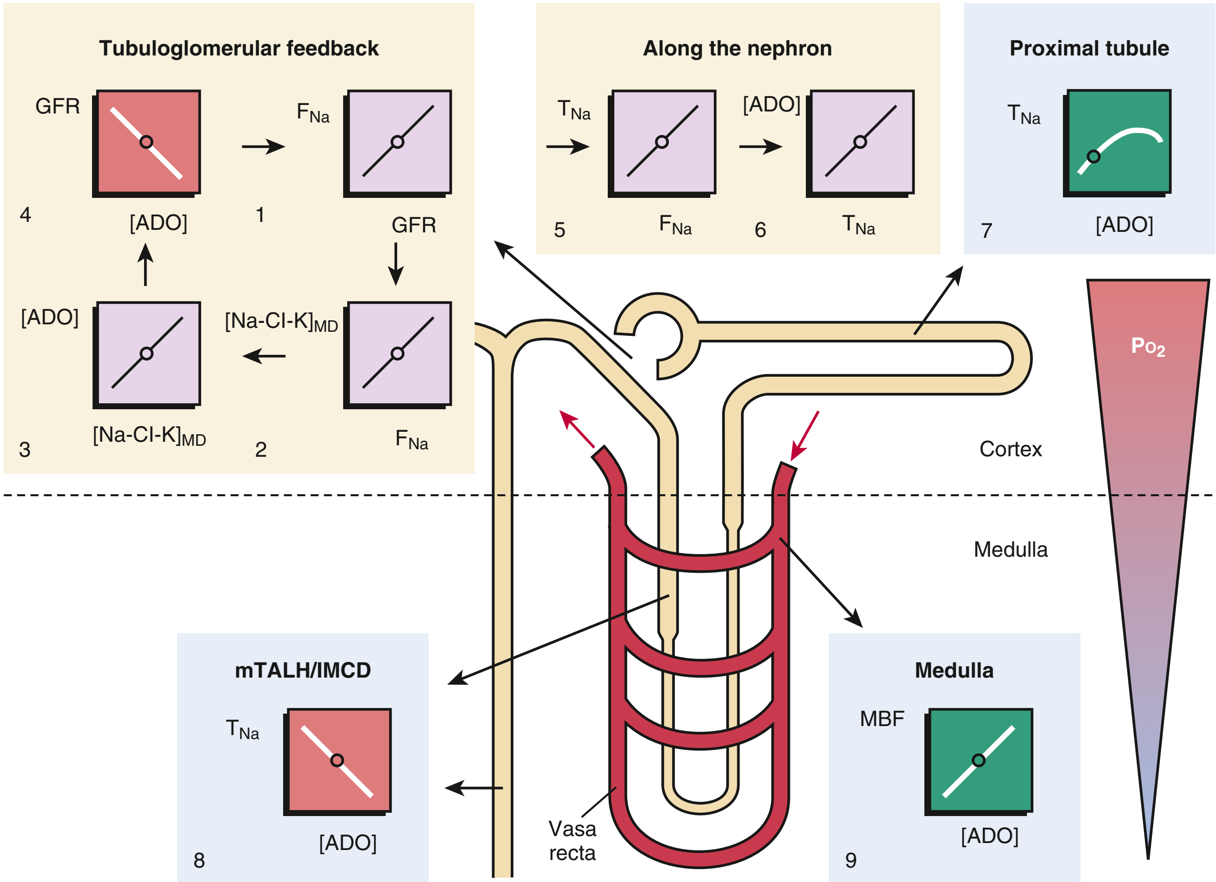

Fig. 5.13 Role of extracellular adenosine (*ADO*) in protecting the renal medulla from hypoxia. The line plots illustrate the relationships between the given parameters. *Small circles* on these lines indicate ambient physiologic conditions. 1. A rise in glomerular filtration rate *(GFR)* increases the Na^+ load *(F_{Na})* to the tubular system in cortex and medulla. 2. This rise in F_{Na} increases the salt concentration sensed by the macula densa (*$[Na\text{-}Cl\text{-}K]_{MD}$*). 3. The increase in $[Na\text{-}Cl\text{-}K]_{MD}$, in turn, enhances local ADO. 4. ADO lowers GFR and thus F_{Na}, which closes a negative feedback loop and thus provides a basis for an oscillating system. 5. F_{Na} determines Na^+ transport work *(T_{Na})* and O_2 consumption in every nephron segment, and thus oscillations in F_{Na} may help protect the medulla. 6. A rise in T_{Na} increases ADO along the nephron. 7. In the cortical proximal tubule, ADO stimulates T_{Na} and thus lowers the Na^+ load to segments residing in the medulla. 8. In contrast, ADO inhibits transport work in the medulla including medullary thick ascending limb *(mTAL)* and inner medullary collecting duct *(IMCD)*. 9. In addition, ADO enhances medullary blood flow *(MBF),* which increases O_2 delivery and further limits O_2-consuming transport in the medulla. (Modified from Vallon V, Muhlbauer B, Oswald H. Adenosine and kidney function. *Physiol Rev*. 2006;86:901–940.)

are subject to regulation. Dissociating RBF from GFR equates to changing the glomerular filtration fraction. This can work to stabilize kidney O_2 because lowering the filtration fraction increases the ratio of supply to demand for O_2. For nephrons in filtration equilibrium, this requires independent control of the afferent and efferent arterioles, which can be achieved by modulating relative activities of purinergic, angiotensin, nitric oxide, and other signaling systems in the glomerulus. A full discussion of glomerular hemodynamics is available in Chapter 3, but a few features are noted here.

To begin, filtration fraction can be lowered by constricting the afferent arteriole (which reduces net O_2 delivery) or dilating the efferent arteriole (which increases net O_2 delivery). Constricting the afferent arteriole confers initial energy savings by reducing GFR faster than RBF, but there is diminishing return as O_2 delivery declines toward the basal requirement. Dilating the efferent arteriole reduces GFR only when glomerular capillary pressure is low to begin with, such as during hypotension or with high afferent resistance.[99] When angiotensin II acts on the afferent and efferent arterioles to stabilize GFR in the face of low blood pressure or high upstream resistance by preferentially constricting efferent arterioles, the kidney is accepting a decrease in the ratio of O_2 supply to O_2 demand. Conversely, adenosine signaling in the glomerulus decreases filtration fraction and so manages to stabilize nephron function without compromising the O_2 supply–demand balance. Adenosine in the nanomolar range constricts the afferent arteriole via high-affinity A_1 adenosine receptors. Higher adenosine concentration dilates the efferent arteriole via low-affinity A_2 adenosine receptors. Interstitial adenosine concentration rises as more NaCl is delivered into the nephron. The prototype for this is the tubuloglomerular feedback (TGF) signaling through the macula densa, although other sources are not precluded (Fig. 5.13). When the kidney is operating in the domain of modest distal delivery, increasing the TGF signal constricts the afferent arteriole. When the kidney is operating in the domain of high distal delivery, further increase causes the efferent arteriole to dilate,[100] which can be viewed as a shift in priority toward maintaining the O_2 supply as the supply diminishes.

The second generic means for stabilizing kidney O_2 is to alter QO_2/T_{Na}. It is theoretically possible to alter QO_2/T_{Na} in a number of ways.[101,102] Transport could be shifted from the proximal tubule, where efficient use of energy from the Na^+-K^+-ATPase drives passive transport, to other segments where all Na^+ reabsorbed passes through the Na^+-K^+-ATPase. Tubular backleak permeability could change, which would affect the number of times that a given Na^+ ion must be reabsorbed to escape excretion into the urine. The ratio of ATP produced per O_2 consumed could be altered by the regulated activity of UCPs (see Fig. 5.3).[7] Finally, ATP could be diverted to gluconeogenesis and other basal metabolic needs unrelated to tubular transport. It can be challenging to estimate the basal metabolic activity of the kidney, which may change under different physiologic conditions such as greater gluconeogenesis during fasting.

The same neurohumoral factors that exert well-known effects on glomerular hemodynamics and O_2 supply, including nitric oxide, angiotensin II, adenosine, and catecholamines, also appear to participate in the regulation of kidney metabolism and QO_2 by the tubule. It has been shown[103,104] that the administration of nonselective nitric oxide synthase (NOS) inhibitors increases QO_2/T_{Na}. Other experiments suggested that NOS-1 is the specific isoform that regulates this action in vivo.[103] The changes in QO_2 with NOS inhibition may occur due to 1. a shift in the site of sodium reabsorption to a less efficient nephron segment; 2. decreased efficiency of transport in the proximal tubule (i.e., decrease in the passive component of reabsorption); or 3. less efficient use of O_2 by mitochondria. For example, nitric oxide given directly to a proximal tubular cell is both a proximal diuretic[105] and a competitive inhibitor of O_2 flux through the ETC in mitochondria.[106] Most effects of nitric oxide are mediated by cyclic guanosine monophosphate, but the mitochondrial effect is presumed to occur through the competitive inhibition of cytochrome *c* oxidase.[106-108]

Rats and mice with angiotensin-induced hypertension exhibit stimulation of sodium transporters from the cortical TAL to the CD, in regions with higher QO_2/T_{Na}, and inhibition or no stimulation of sodium transporters in the proximal nephron, where QO_2/T_{Na} is lower.[109,110] Studies in spontaneous hypertensive rats have suggested opposing effects of angiotensin II and nitric oxide on the QO_2/T_{Na} ratio in the kidney.[111] Rats with angiotensin-induced hypertension demonstrated an increased QO_2/T_{Na} that was reversed by a mimetic of superoxide dismutase, which is consistent with the theory that many angiotensin II effects are mediated by upregulating the activity of reduced nicotinamide adenine dinucleotide phosphate (NADPH) oxidase.[112] In addition, there is evidence that angiotensin II contributes to mitochondrial dysfunction and oxygen consumption in aging rats.[113] The role of angiotensin II in kidney metabolism is implicated in the ablation/infarction remnant kidney model of chronic kidney disease (CKD). Oxygen consumption factored for nephron number or T_{Na} has been shown to be elevated in this model[114-116] and lowered by various treatments including angiotensin blockade.[114]

REGULATORY ROLE OF KIDNEY METABOLISM

The functional significance of intermediary metabolism is not limited to bioenergetics. It is now well recognized that many metabolic intermediaries regulate cellular function through their effects on gene expression, signal transduction, and other mechanisms. In the kidney, intermediary metabolism may directly influence tubular transport activities by changing the ATP availability but may also regulate physiological functions through mechanisms unrelated or not directly related to the bioenergetics of tubular transport. In some cases, kidney metabolism may influence a regulatory pathway, which, in turn, influences the bioenergetics of tubular transport or feeds back to modulate kidney metabolism. The regulatory role of kidney metabolism influences normal renal physiology, as well as the development of hypertension, acute kidney injury, and CKD.

TUBULOGLOMERULAR FEEDBACK

Significant fluctuations in RBF, GFR, and filtered Na^+ load would overwhelm the kidney's ability to accurately match Na^+ and volume output to input and compromise homeostasis of extracellular fluid volume. This does not normally occur because RBF and GFR are tightly controlled by the TGF mechanism (described in detail in Chapter 3). In short, if RBF and/or GFR increases and GTB maintains a constant fractional reabsorption along the proximal tubule, an increasing amount of salt will be delivered to the macula densa, which sets off the TGF response. Specifically, increases in apical NaCl delivery or flow to this region provoke the cells of the macula densa to release ATP into the interstitium surrounding the afferent arterioles. This response is dependent on the basolateral Na^+-K^+-ATPase to maintain the inward-directed Na^+ gradient.[117] ATP release is via maxi-anion channels.[118] Some fraction of the released ATP is converted to adenosine by local ectonucleoside triphosphate diphosphohydrolase 1 (ecto-NTPDase1) and ecto-5′-nucleotidase.[119] This adenosine activates A_1 adenosine receptors on the afferent arteriole, causing vasoconstriction. The arteriolar constriction reduces RBF and GFR in concert until Na^+ delivery to the macula densa is realigned. Thus an inverse relationship is established between tubular NaCl load and the GFR of the same nephron.[20]

Due to the time it takes for information to pass through the TGF system, the system is prone to oscillate with a period of around 30 seconds. Rhythmic oscillations of kidney PO_2 occur at the same frequency as TGF-mediated oscillations in tubular flow. This illustrates the simultaneous influence of TGF over minute-to-minute tubular flow rate and oxygen levels in the kidney.[120]

Adenosine mediates TGF as a vasoconstrictor. Adenosine-mediated vasoconstriction is unique to the afferent arteriole. In all other beds where adenosine is vasoactive, it exerts a vasodilatory effect mediated by A_2 receptors. In addition to adenosine receptors, the afferent arteriole expresses P_{2X} purinergic receptors that also mediate a vasoconstrictor response, in this case to interstitial ATP. These P_{2X} receptors are essential to pressure-mediated RBF autoregulation,[121] but adenosine A_1 receptors are sufficient to explain the TGF response.[119]

Adenosine also plays an important role in stabilizing medullary energy balance through local adjustments in blood flow and transport along with other autocrine and paracrine factors, including vasodilatory prostaglandins and nitric oxide, which increase medullary blood flow (MBF) while inhibiting sodium transport in the TAL.[33,122,123] Adenosine, in particular, is a case study in local metabolic regulation by negative feedback in the medulla. When ATP levels decline, adenosine is released from TAL cells into the renal interstitium, where it binds to adenosine A_1 receptors and inhibits Na^+ reabsorption in the TAL and IMCD. This has the effect of increasing Po_2 by reducing Qo_2. The same pool of adenosine also activates vascular adenosine A_2 receptors in the deep cortex and medullary vasa recta to increase blood flow.[124,125]

By these mechanisms, the TAL looks after its own interest. But because TAL sodium reabsorption normally exceeds the urinary sodium excretion by 40-fold, any significant decline in TAL reabsorption must be compensated for by increasing active transport somewhere else or by reducing GFR through TGF. Activation of A_1 receptors in the glomerulus, proximal tubule, or TAL contributes to lessening the amount of work imposed on the hypoxic outer medulla, whereas activating A_2 receptors in the vasa recta supports O_2 delivery to the medulla (summarized in Fig. 5.13).

REACTIVE OXYGEN SPECIES

Changes in renal oxygen metabolism and mitochondrial bioenergetics may lead to changes in reactive oxygen species (ROS) production. ROS may be produced by mitochondria, NAD(P)H oxidase, and other sources. In mitochondria, electron "leaks" from the ETC may lead to a one-electron reduction of O_2 and the generation of superoxide. Complexes I and III in the ETC are the primary sources of mitochondrial superoxide.[126,127] Other sites of ROS formation in mitochondria may include α-ketoglutarate dehydrogenase, pyruvate dehydrogenase, glycerol 3-phosphate dehydrogenase, and fatty acid β oxidation.[128] Excessive ROS, particularly superoxide and hydrogen peroxide, is present in the kidney of animal models of kidney disease and hypertension and may contribute to the development of disease through several mechanisms, such as decreasing NO bioavailability.[129,130]

Uncoupling proteins (UCP) allow proton leakage back across the inner mitochondrial membrane without generating ATP and could lower oxygen utilization for ATP production and increase oxygen consumption. UCP-2 and UCP-3 may reduce mitochondrial ROS generation while the effect of UCP-1 is more complicated.[131] UCP-$2^{-/-}$ mice exhibit increased superoxide production, decreased NO bioavailability, and hypertension with a high-salt diet for 24 weeks.[132] Mice null for the redox-sensitive chaperone DJ-1 exhibit hypertension and an upregulation of renal UCP2 expression. UCP2 knockdown by renal subcapsular infusion of a siRNA attenuates hypertension and increases serum NO metabolite levels in these mice.[133]

TCA CYCLE INTERMEDIATES

An endogenous insufficiency of fumarase has been shown to contribute to the development of hypertension in Dahl salt-sensitive rats.[134,135] Consistent with the role of the kidney in this effect, kidney-specific knockdown of fumarase increases the blood pressure salt sensitivity of Sprague-Dawley rats.[136] Fumarase converts fumarate to L-malate in the TCA cycle. L-malate is converted to oxaloacetate. Oxaloacetate may be combined with acetyl-CoA to form citrate in the TCA cycle but also could be converted to aspartate via aspartate transaminase. Aspartate may be combined with citrulline by argininosuccinate synthase to form argininosuccinate, which is converted to L-arginine and fumarate by argininosuccinate lyase. L-arginine is the substrate for NO synthase (NOS) for the generation of NO and citrulline. Renal NO protects against the development of hypertension through its vasodilatory effect, as well as direct inhibition of sodium reabsorption at several nephron segments.[137,138] L-arginine, given systematically or directly into the renal interstitium, substantially attenuates hypertension in SS rats.[139,140] As such, fumarase insufficiency in SS rats might contribute to salt-induced hypertension by decreasing L-arginine and NO in the kidney.[141] A heterozygous mutation in Nos3 that results in haploinsufficiency of eNOS exacerbates hypertension and renal injury in SS rats. Transgenic overexpression of fumarase in these rats increases NO and the ratio of L-arginine/citrulline in the renal outer medulla and abolishes hypertension and renal injury that may be attributed to Nos3 heterozygous mutation.[142]

A connection has also been established between local accumulation of the TCA cycle intermediate succinate and activation of the RAS.[140] Intravenous injection of succinate into rats and mice induces hypertension via activation of the renin-angiotensin system, and this response is abolished in GPR91-deficient mice.[143] The activation of succinate receptor, GPR91, could trigger renin release from macula densa cells in the distal tubule.[144,145] Succinate can accumulate extracellularly when oxygen supply does not match demand. In the extracellular fluid, it can bind to its G protein–coupled receptor, GPR91. PO_2 in the juxtaglomerular region is reduced in the hyperglycemia of diabetes, and succinate levels are high in urine and renal tissue of diabetic animals. Inhibition of the TCA cycle's succinate dehydrogenase complex causes robust renin release. This effect is amplified in high-glucose conditions or with added succinate. In summary, GPR91-mediated signaling in the juxtaglomerular apparatus could modulate GFR and RAS activity in response to changes in metabolism (especially after a meal when glucose level is elevated). Pathologically, GPR91-mediated signaling could link metabolic diseases (such as diabetes) with RAS activation, systemic hypertension, and organ injury.

DNA methylation and demethylation in the kidney are involved in the development of hypertension and kidney disease.[146-148] DNA demethylation catalyzed by ten-eleven translocase requires the TCA cycle intermediate, α-ketoglutarate. The role of α-ketoglutarate in the regulation of DNA methylation in the kidney and development of hypertension and kidney disease remains to be investigated.

AMINO ACID METABOLISM

Renal metabolism of several amino acids may contribute to the regulation of kidney function. The relation of these amino acids with renal energy metabolism is largely unclear because amino acids, except for glutamine, are not normally a key source of energy in the kidneys. However, it is possible that amino acids are used as fuel in the kidneys when renal metabolic abnormalities occur. A combined treatment of a high-salt diet with saline drinking in mice causes broad changes in energy and substrate metabolism in the liver and skeletal muscle, including amino acid catabolism in the muscle, in an apparent attempt to support urea production and water conservation.[149]

The antihypertensive effect of L-arginine, likely through enhancing NO production in several tissues including the kidney, is well established in animal models. L-arginine, administered through renal medullary interstitial infusion[139] or other routes, increases the generation of NO and substantially attenuates hypertension in SS rats. Renal L-arginine may come from the endogenous synthesis in the kidneys or circulating L-arginine. Circulating L-arginine mainly comes from intestinal absorption of protein-derived L-arginine and free L-arginine in the food.[150] Endogenous L-arginine is mainly synthesized in the liver and kidneys through the urea cycle. L-arginine synthesized in the liver does not reach the systemic circulation effectively because of the high activity of hepatic arginase.[151,152] L-arginine transport, which may be inhibited competitively by L-lysine, also appears to be involved in angiotensin II–induced renal cortical vasoconstriction in SD rats and low renal NO bioavailability in SHR.[153,154]

Citrulline and aspartate are the substrates of endogenous L-arginine synthesis in the kidney. Citrulline is a nonessential nonprotein amino acid mainly derived from intestinal glutamine breakdown. The liver does not take up citrulline.[155-157] The kidney may take up circulating citrulline and convert it to L-arginine. Argininosuccinate synthase is a rate-limiting enzyme in the citrulline-NO cycle, and its expression and activity can be induced by citrulline.[158] Citrulline is effective in attenuating hypertension.[140,159,160]

The metabolism of glycine, glutamate, and cysteine may influence the homeostasis of glutathione (GSH) and glutathione disulfide (GSSG). The synthesis of GSH, an important antioxidant, from glycine, glutamate, and cysteine is catalyzed by GSH synthetase and glutamylcysteine synthetase and regulated by cysteine availability and GSH/GSSG feedback inhibition.[161] Glutathione reductase and peroxidase catalyze GSH and GSSG interconversion. Cysteine, delivered as its stable analog N-acetyl cysteine, has antihypertensive effects in humans and animal models and may work directly or through its storage form GSH to decrease oxidative stress.[162]

Catecholamines including dopamine, norepinephrine, and epinephrine play a significant role in regulating renal hemodynamics, renal tubular transport, and blood pressure. Catecholamines are metabolic products of the amino acid tyrosine. Renal proximal tubules lack tyrosine hydrolase required for converting tyrosine to 3,4-dihydroxyphenylalanine (L-DOPA) and dopamine β-hydroxylase required for converting dopamine to norepinephrine and epinephrine. However, renal proximal tubules and possibly the distal nephron may take up L-DOPA and convert it to dopamine.[163]

Urinary levels of β-aminoisobutyric acid (BAIBA), a nonprotein amino acid produced by catabolic metabolism of thymine or branched-chain amino acid valine, are inversely correlated with systolic blood pressure in humans on low- and high-sodium intakes.[164] Treatment with BAIBA significantly attenuates salt-induced hypertension in SS rats.[164] Alanine-glyoxylate aminotransferase-2 (AGXT2) is one of the enzymes involved in the metabolism of BAIBA. AGXT2 also may degrade asymmetric dimethylarginine, an endogenous inhibitor of NOS. AGXT2 knockout mice exhibit increased asymmetric dimethylarginine and reduced NO and develop hypertension.[165] Treatment of SS rats with a high-salt diet downregulates valine and another branched-chain amino acid leucine in glomeruli.[166]

ACUTE KIDNEY INJURY AND CHRONIC KIDNEY DISEASE

Alterations of kidney metabolism occur broadly in acute kidney injury and CKD and may play an important role in the development of these diseases, which are discussed in detail in other chapters. Several examples are described briefly here.

Studies in ischemia-reperfusion model in rats and pigs demonstrated reduced oxygenation and persistent tissue hypoxia in the early stages after reperfusion (3–4 hours post ischemia),[167,168] which was more prominent in the outer medullary regions. In postcardiac surgery patients with or without AKI, increased renal O_2 extraction and Q_{O_2}/T_{Na} were demonstrated in the AKI group.[169] Improvement in renal oxygenation with loop diuretics due to reduction in tubular transport–related Q_{O_2} was also demonstrated.[170]

Intrarenal hypoxia has been proposed as a final common pathway to progression of CKD.[171] In late stages of CKD, rarefaction of capillaries and other structural changes have been implicated in the decrease in oxygen supply leading to hypoxia. However, intrarenal hypoxia has been demonstrated in the early stages before any structural changes.[172] High Q_{O_2}/T_{Na} has been postulated to be the etiology of tubular hypoxia in the early stages of CKD.[114] In early experimental diabetes, decreased tissue oxygen tension (P_{O_2}) has also been demonstrated before any structural changes associated with diabetic nephropathy.[173]

In mouse models of autosomal dominant polycystic kidney disease (ADPKD) and in kidney tissue from patients with ADPKD, defective glucose metabolism—specifically, increased aerobic glycolysis—was observed.[174] Moreover, treatment with 2-deoxyglucose, an inhibitor of glycolysis, lowered kidney weight, volume, cystic index, and proliferation rates in mouse models of ADPKD.[175] This phenotypic feature of enhanced aerobic glycolysis, also known as Warburg effect, is typically seen in proliferating cancer cells. In these mice models of *Pkd1* gene inactivation, there was inhibition of liver kinase B1 (LKB1)-AMPK axis along with activation of mTOR complex I pathway, leading to increased glycolysis.[174]

Kang and colleagues[176] described reduced levels of enzymes and regulators involved in fatty acid oxidation in mouse models of renal fibrosis. In transcriptional analyses of microdissected human kidney samples from patients with diabetic or hypertensive CKD, changes in fatty acid metabolism, β-oxidation, amino acid catabolism, and carbohydrate metabolism were observed. Genes related to fatty acid metabolism and their key transcriptional regulator complex, PPARA–PPARGC1A, were markedly lower in CKD samples compared with samples obtained from people with normal kidney function. This was associated with higher lipid accumulation in renal tubular epithelial cells. Key regulators of glucose utilization were also lower in CKD samples. Using elegant experiments with genetic and pharmacologic approaches, the reduction in fatty acid oxidation was shown to be directly implicated in the pathogenesis of renal fibrosis in CKD. Although lipid accumulation in tubular cells in CKD patients has been reported before, these findings showed that it is the reduction in fatty acid oxidation rather than just the accumulation of lipid in tubular cells that plays a role in the development of renal fibrosis.

Sas and colleagues[177] provided a comprehensive assessment of substrate metabolism in diabetes using a systems-based approach that included transcriptomics, metabolomics, and metabolic flux analyses to determine alterations in glucose and fatty acid metabolism. In the kidney cortex in a type 2 diabetic mouse model (db/db mice), transcriptomic and metabolomic profiling demonstrated an increase in glycolysis, fatty acid β-oxidation, and TCA cycle flux. Increased renal metabolism was associated with increased protein acetylation, which is a nutrient-sensing posttranslational modification. There was also evidence for mitochondrial dysfunction. Remarkably, transcriptomic analysis of kidney biopsy samples in patients with type 2 diabetes identified significant enrichment of many pathways involved in fatty acid, glucose, and amino acid metabolism, and network analysis showed results similar to those obtained in db/db mice. In another cohort of type 1 diabetic patients enrolled in the Finnish Diabetic Nephropathy (FinnDiane) Study, increased glycolytic (hexose-6-phosphates, 2,3-phosphoglycerate) and TCA cycle metabolites (succinate, fumarate, and malate) were detected in the urine compared with healthy control subjects. Urine samples from a subset of patients from the Family Investigation of Nephropathy and Diabetes (FIND) study showed increased glycolytic intermediates at baseline in diabetic subjects compared with control subjects. TCA cycle intermediates in the urine were also elevated in diabetic subjects and predicted diabetic kidney disease progression, demonstrating their potential as prognostic biomarkers of diabetic kidney disease progression. These and previously discussed studies highlight the adaptive and maladaptive metabolic reprogramming and its role in the pathophysiology of diabetic and nondiabetic CKD.

MITOCHONDRIAL DISORDERS

The mitochondrial genome is distinct from the nuclear genome, and it encodes 13 of the 88 protein subunits of ETC complexes I through V, as well as 22 mitochondrial-specific transfer RNAs (tRNAs) and two RNA components of the translational apparatus. The nuclear genome encodes the remaining respiratory chain subunits and most of the mitochondrial DNA replication and expression components.

Several lines of investigation support the significant role of early mitochondrial dysfunction in the pathophysiology of acute and CKD. Disorders affecting mitochondrial oxidative phosphorylation can arise from mutations in either mitochondrial genes or nuclear genes encoding respiratory chain components or ancillary factors involved in maintenance of ETC function or the overall number of mitochondria.[178] The incidence of genetic mitochondrial disorders is estimated at about 1 in 5000 births, with the most common affecting the sequence of a mitochondrial tRNA for leucine. Such mutations affect mitochondrial function in all tissues. Symptoms are evident before 2 months of age, and the number of organ systems affected increases with age. A homoplasmic mutation substituting cytidine for uridine immediately 5′ to the mitochondrial tRNAIle anticodon causes a cluster of maternally inherited diseases including hypertension in a large kindred.[179] Other mutations in mitochondrial tRNAs also reportedly cause maternally inherited hypertension, and these mutations decrease the efficiency of mitochondrial oxygen utilization.[180]

Impairment of mitochondrial oxidative phosphorylation results in increased levels of reducing equivalents (NADH, FADH2), which, in the mitochondria, transform acetoacetate to 3-hydroxybutyrate and, in the cytosol, transform pyruvate to lactate. Thus elevated levels of lactic acid, ketone bodies, and impaired redox status are suggestive of a mitochondrial defect disorder.[181] If the genetic cause of the impairment can be pinpointed, then an appropriate therapy, if available, can be implemented to treat these life-threatening disorders; for example, coenzyme Q_{10} enzyme defects can be treated with coenzyme Q_{10} supplementation.[178]

Myopathies and cardiomyopathies are the most common manifestations of mitochondrial disease and central nervous system symptoms including encephalopathies are common. Renal system impairment can be present but is not seen without other system deficiencies and is usually reported in children. Although glomerular disease and tubulointerstitial nephropathy have both been reported, the most frequently observed is impairment of proximal tubule reabsorption, known as de Toni–Debré–Fanconi syndrome, in which there are urinary losses of bicarbonate, amino acids, glucose, phosphate, uric acid, potassium, and water. All symptoms can be explained by a lack of ATP to fuel Na^+-K^+-ATPase sufficiently to drive transepithelial transport, primarily in the proximal tubule. The symptoms can range from mild to more severe and present in the neonatal period in most patients. Biopsy specimens show tubular dilations, casts, dedifferentiation, and cellular vacuolization. At the cellular level, there are enlarged mitochondria. Supplements of sodium bicarbonate, potassium, vitamin D, phosphorus, and water are called for if these symptoms are evident.[178,181]

Significant changes in mitochondrial function in diabetic kidneys have been described and recently reviewed in detail.[4,5] Early changes in mitochondrial energetics, increased mitochondrial fragmentation, and reduced levels of PGC1a have been described in diabetic kidneys. In diabetic nephropathy, increased metabolism has been shown to be associated with mitochondrial dysfunction.[177] Assessment of oxidative phosphorylation in mitochondria isolated from renal cortexes of 24-week-old control and diabetic mice showed increased proton leak and diminished mitochondrial ATP production. Additionally, total mitochondrial capacity was decreased in diabetic mitochondria, and the expression of mitochondrial uncoupling protein 2 was increased fourfold. In renal cortexes from diabetic mice, expression of proteins in complexes I, II, III, and complex IV subunit cytochrome *c* oxidase subunit 4 (COX4) were significantly decreased at 24 weeks. The results of these studies suggested that dysfunction of ETC proteins in mitochondria from diabetic kidneys leads to less efficient ATP production and compensatory increases in glucose and fatty acid metabolic flux. This may be the primary metabolic abnormality in the diabetic kidney cortex as it precedes other markers of renal injury. Another study compared urine metabolites in patients with diabetes, but with or without CKD and in healthy controls.[182] Bioinformatic analyses revealed that 12 of the 13 differential metabolites were linked to mitochondrial metabolism. Kidney sections in patients with diabetes and CKD showed lower expression of mitochondrial proteins and lower gene expression of PGC1a, which is a major regulator of mitochondrial biogenesis. Urine exosomes in these patients also showed less mitochondrial DNA. Evidence of mitochondrial dysfunction in nondiabetic CKD is also emerging. Early changes in mitochondrial function and structure in animal models of CKD have been reported.[183] In patients with CKD, mRNA levels of several mitochondrial enzymes and transcription factors were found to be lower, although no differences in mitochondrial copy number were seen.[176] In another clinical study, mitochondrial DNA copy number was evaluated in peripheral blood in a population cohort.[184] Participants with higher mitochondrial DNA copy numbers had a lower rate of prevalent diabetes and lower risk of incident CKD, even after adjusting for various risk factors for CKD.

The role of mitochondrial dysfunction in AKI has received significant attention. Several publications have elucidated the underlying mechanisms of mitochondrial dysfunction in various etiologies of AKI (reviewed in detail[4]). In ischemic and nephrotoxic AKI, decreased mitochondrial mass, disruption of cristae, and significant mitochondrial swelling have been observed.[185] In ischemic AKI, increase in mitochondrial NADH and dissipation of mitochondrial membrane potential in proximal tubules has been demonstrated.[186] In ischemic and myoglobulinuric AKI, reduction in various ETC proteins in the proximal tubule was shown.[187] Other studies have uncovered the crucial role of mitochondrial biogenesis using stimulators of PGC1α, master regulator of mitochondrial biogenesis in recovery from ischemic AKI[188] and sepsis-associated AKI.[189] The role of mitochondrial dynamics has also been elucidated in some forms of AKI. In an elegant study, Brooks and colleagues[185] described the disruption of mitochondrial dynamics and its pathogenic role in ischemic and nephrotoxic AKI.[185] Mitochondrial fragmentation was shown to precede tubular cell injury and death, and pharmacologic inhibition of Drp1 prevented fragmentation and ameliorated AKI. Further highlighting the role of mitochondrial dysfunction, several pharmacologic targets to improve mitochondrial function in AKI and CKD have been investigated and found to be effective.[190]

COMMON GENETIC VARIATIONS

Common DNA sequence variations in humans, such as single nucleotide polymorphisms (SNPs), may influence intermediary metabolism and mitochondrial function, which, in turn, influence renal function or related phenotypic traits such as blood pressure. For example, several genes known to be involved in intermediary metabolism or mitochondrial function contain nonsynonymous and potentially damaging SNPs that are associated with human blood pressure.[3] An expression quantitative trait locus (eQTL) is a DNA sequence variant for which individuals possessing different alleles of the variant show different expression levels of a gene in one or more tissues. Several genes known to be involved in intermediary metabolism or mitochondrial function are eQTL genes for noncoding SNPs associated with human blood pressure.[3] The specific role of the kidney in mediating the effect of these DNA sequence variations on blood pressure remains to be examined.

HYPOXIA-INDUCIBLE FACTOR

One of the major transcription factors mediating the cellular adaptation to hypoxia is the oxygen-sensitive, hypoxia-inducible factor (HIF).[191] A great body of work by Semenza and colleagues[191-194] has described the role of HIF as a primary oxygen sensor and regulator of cellular oxygen homeostasis. It accumulates in hypoxic cells, where it acts to regulate gene expression. HIF consists of a labile α subunit (HIF1α, HIF2α, HIF3α) and a constitutive β subunit. These subunits heterodimerize to form a transcriptional complex that translocates to the nucleus and binds to hypoxia response elements of various hypoxia-responsive genes.[195,196] HIF1α and HIF2α have been well studied, have similar structures, and have significant overlap in their actions on target genes; however, some target genes appear to be exclusively under the regulation of one or the other. Their renal tissue expression patterns also differ, with predominant expression of HIF1α in the tubular epithelial cells and HIF2α in the interstitial fibroblasts and peritubular endothelial cells in the hypoxic kidney.[197,198] There is limited information regarding the function and actions of HIF-3α.

During adaptation to hypoxia, HIF1α and 2α regulate the expression of many genes that regulate oxygen delivery and consumption. Changes that culminate in a rise in erythropoiesis, vasodilation, and tissue vascularization all increase oxygen delivery.[199,200] Another set of responses conserve energy by decreasing substrate movement into the tricarboxylic acid cycle, increasing cellular glucose uptake and glycolytic enzymes to increase anaerobic ATP production, and shifting the cell toward glycolytic metabolism.[194] HIF-1α also has significant effects on mitochondrial metabolism; specifically, it diminishes NADH supply to the ETC, induces a subunit switch in complex IV of ETC to optimize its efficiency in hypoxia, and represses mitochondrial biogenesis and respiration.[201,202] Finally, HIF-1α also induces mitochondrial autophagy as an adaptive metabolic response to prevent increased levels of ROS generation and cell death in hypoxia.[203]

Recently, HIF-1α has also been recognized as a regulator of salt transport. High-salt intake increased HIF-1α expression in the renal medulla.[204] Inhibition of HIF-1α in the renal medulla decreased MBF, blunted urine flow, and urinary Na excretion. In the presence of HIF-1α inhibitor, rats on a high-salt intake developed positive cumulative salt balance and higher blood pressure. Thus medullary HIF-1α inhibition on a high-salt intake leads to resetting of pressure natriuresis, Na retention, and salt-sensitive hypertension.[205] HIF-1α expression has been reported in the medulla in a normal rat,[206] where selective inhibition of medullary HIF-1 induced significant tubulointerstitial damage. Interestingly, HIF-1 expression appeared to correlate with salt transport—an increase noted after an increase in medullary workload and a decrease in expression noted after inhibiting Na transport in the TAL by furosemide administration, which also increased medullary Po_2.[206]

HIF activity is regulated by proteasomal degradation during normoxia by von Hippel-Lindau-E3 ubiquitin ligase complex after being hydroxylated by prolyl 4 hydroxylase domains (PHDs). Of the three main identified PHDs (1, 2, and 3), PHD2 is the main enzyme that targets HIF for degradation under normoxia.[197] All three PHDs are expressed in the kidney and are found predominantly in the DCT, collecting duct, and podocytes, and levels are depressed during ischemia and reperfusion.[196,197] HIF activity is also regulated by factor inhibiting HIF (FIH), which inhibits its transcriptional activity by hydroxylating an asparagine residue within the transactivation domain and prevents the binding of coactivators to the HIF transcriptional complex.[194]

A tremendous amount of progress has been made in understanding how HIF helps to maintain O_2 homeostasis and the study of this factor in the kidney under normal physiologic and pathophysiologic conditions. Particularly, the role of HIF stabilization and activation in the kidney to stimulate erythropoietin production has been targeted therapeutically. Structural analogs of 2-oxyglutarate, which are the substrates used by PHDs for HIF hydroxylation, have been developed to reversibly inhibit PHD activity. Several HIF prolyl hydroxylase inhibitors have been evaluated as a treatment for anemia in patients with CKD in several phase 3, randomized, controlled trials. Initial results in patients with dialysis-dependent CKD are promising, while the indications and safety profiles in patients with non-dialysis-dependent CKD are less clear.[207]

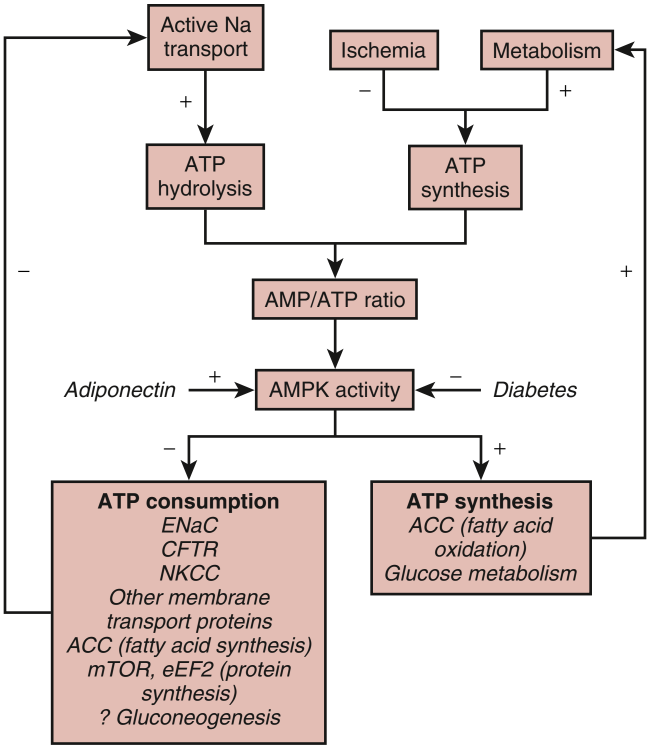

Fig. 5.14 Proposed role of adenosine monophosphate–activated protein kinase *(AMPK)* in the kidney in coupling catabolic pathways requiring ATP hydrolysis (primarily sodium transport) with metabolic pathways leading to ATP synthesis (primarily fatty acid and glucose oxidation). +, Activating pathway; –, inhibitory pathway. *ACC,* acetyl-CoA carboxylase. (From Hallows KR, Mount PF, Pastor-Doler M, Power DA. Role of the energy sensor AMP-activated kinase in renal physiology and disease. *Am J Physiol Renal Physiol*. 2010;298:F1067–F1077.)

ADENOSINE MONOPHOSPHATE–ACTIVATED PROTEIN KINASE

The energy status of the cell can be detected by the ultrasensitive 5′-adenosine monophosphate (AMP)–activated protein kinase (AMPK), which is a ubiquitously expressed, highly conserved, key energy sensor and regulator of cellular metabolic activity.[208] AMPK consists of a heterotrimer of a catalytic subunit ($\alpha 1$ or $\alpha 2$), together with a β ($\beta 1$ or $\beta 2$) and a γ ($\gamma 1$, $\gamma 2$, or γ 3) regulatory subunit.[208] Cellular energy stress, which can be due to a variety of conditions, such as nutrient or glucose deprivation, exercise, hypoxia, or ischemia, is detected as a rising concentration of AMP and an increase in the AMP/ATP ratio. AMPK is activated by phosphorylation of the α catalytic subunit on threonine-172 (Thr172) by upstream kinases.[209] The binding of AMP to the γ-regulatory subunit of AMPK increases its activity in three ways: 1. conformational change in AMPK, which allows enhanced phosphorylation of the α catalytic subunit on Thr172 by upstream kinases, thus activating AMPK; 2. inhibition of dephosphorylation of the catalytic subunit; and 3. direct allosteric activation. These three effects, working in concert, render the system exquisitely sensitive to changes in AMP, and all are antagonized by ATP—thus the importance of the AMP/ATP ratio. AMPK acts as a metabolic checkpoint to facilitate metabolic adaptation to cellular energetic stress by triggering ATP producing pathways, such as fatty acid oxidation, glucose uptake, and glycolysis, while inhibiting ATP-consuming pathways, such as fatty acid synthesis, protein synthesis, and potentially active transport[210] (Fig. 5.14). AMPK also promotes cellular autophagy, an energy-conserving survival mechanism in low-energy states by inhibiting mTOR.[211,212]

AMPK expression in the kidney is seen mainly in the cortical TAL and macula densa cells, as well as in some DCT and collecting ducts.[213] But the understanding of AMPK's impact on energy metabolism and transport in the kidney is just emerging. AMPK has been shown to inhibit the activity of various transport proteins in the lung, gut, and kidney, including ENaC in the kidney collecting duct, Na-K-2Cl cotransporter (NKCC2) in the TAL, and Na^+-K^+-ATPase in the alveolar epithelial cells, particularly during hypoxia (as reviewed[214,215]). Hallows and colleagues have determined that AMPK activation depresses transport mediated by the cystic fibrosis transmembrane conductance regulator, the epithelial sodium channel, the vacuolar H^+-ATPase, and NKCC (see Fig. 5.14). The AMPK pathway may provide a layer of regulation between ATP production by mitochondria and

ATP consumption by transporters. In the Whittam model illustrated in Fig. 5.3, increased active transport provokes a decrease in the ATP/ADP ratio, which drives increased ATP production by the mitochondria. When ATP production by the mitochondria becomes limiting, however, AMPK is likely to be activated, which would drive a reduction in ATP consumption by active membrane transporters. Thus AMPK may regulate the coupling of ion transport and energy metabolism in the kidney.

Regulation of glucose and lipid metabolism by AMPK in pathophysiologic conditions, such as diabetic nephropathy and CKD, has been elucidated in recent publications. In animal models of CKD, diabetes, and obesity, reduced AMPK activity has been described.[216-219] Endogenous AMPK activation by targeting adiponectin (adipose tissue–derived cytokine) and pharmacologic AMPK activation (metformin, AICAR) has been shown to improve glucose and lipid homeostasis in diabetes and obesity-associated kidney disease. In a recent paper, significant changes in renal metabolism with the deletion of a major upstream AMPK activator, liver kinase B1 (LKB1), in kidney distal tubules were observed.[220] Expression of AMPK and other key regulators of metabolism were significantly diminished and accompanied by tubular epithelial injury and interstitial fibrosis. In cultured epithelial cells, loss of LKB1 was associated with diminished fatty acid oxidation and glycolysis, leading to energy depletion and apoptotic cell death. In human kidney samples, lower levels of phosphorylated LKB1 and AMPK α2 subunit were seen in CKD patients. Thus the important role of AMPK in regulating kidney metabolism is being increasingly recognized.

SGLT2 INHIBITION

The full spectrum of the effects of SGLT2 inhibitors and mechanisms involved are covered in Chapters 41 and 54. The renal metabolism–related aspects of SGLT2 inhibition are summarized here. Inhibition of SGLT2 shifts some of the sodium reabsorption from the energy-efficient S1 and S2 segments of the proximal tubule to later parts of the nephron including the S3 segment of the proximal tubule and the energy-expensive medullary TAL. The S3 segment and the medullary TAL are in the outer medulla region, which already has low O_2 availability. The shift of sodium reabsorption increases the energy demand and oxygen consumption, which may stimulate HIF and erythropoietin expression and release and improve oxygenation of various organs.[221] Increased hemoglobin and hematocrit have been shown to explain a large part of the beneficial effect of SGLT2 inhibition on the risk of cardiovascular death.[222] In addition, the increased delivery of sodium to the macula densa may activate TGF via adenosine as described in earlier sections of this chapter, which may explain the reversible reduction of GFR observed following SGLT2 inhibition.[221]

SUMMARY

The kidney has one of the highest specific metabolic rates among all organs. At the level of the whole organ, the major sources of energy for the kidney may include lactate, fatty acids, and glutamine. In vitro and indirect analyses suggest that different nephron segments might prefer different substrates for ATP production, but much remains to be learned about substrate utilization and metabolic activities in specific nephron segments and cell types in the kidney in vivo. The distinct metabolic features of nephron segments may be in part linked to the gradual decrease of tissue oxygenation level from the cortex to the inner medulla. Most of the energy consumed by the kidney is used to fuel the reabsorption of sodium and, secondarily, other solutes. Changes in tubular transport activity are major determinants of renal metabolic activities and oxygenation, and renal metabolism and oxygenation influence tubular transport activities. Although all sodium reabsorption is linked to Na^+-K^+-ATPase, a greater efficiency of sodium reabsorption relative to oxygen consumption is achieved in some nephron segments, especially the proximal tubule, by leveraging passive reabsorption of sodium. It is increasingly recognized that, in addition to its fueling function, renal metabolism may have important regulatory functions where metabolic intermediaries or activities regulate signaling pathways and gene expression that are not directly related to bioenergetics. Renal metabolism may contribute to the regulation of renal physiologic function and the development of kidney-related diseases by influencing bioenergetics and cellular regulatory pathways.

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